LARGE BIPARTITE GRAPHS OF GIVEN DEGREE AND DIAMETER FROM GENERALIZED POLYGONS

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ABSTRACT. Let $n_b(\Delta, D)$ be the largest order of a bipartite graph of maximum degree at most Δ and diameter at most D. We give constructions from generalized polygons that improve the lower bounds in the 2013 table of Miller and Širáň at diameters 3, 4, 6 and 7, and Haar graphs that improve it at diameters 5, 8, 9 and 10. At diameter 3 we add a fourth hat to the three hats of Araujo-Pardo and López and obtain $n_b(q+2,3) \geq 2(q^2+q+6)$ with regular graphs. At diameter 4 we copy points and lines of a generalized quadrangle with weights and stand-ins; a weighted 2-ovoid of the Ahrens–Szekeres quadrangle built from two elliptic quadrics gives $n_b(q+2,4) \geq 2q^2(q+2)$ for odd q, and Singer 2-ovoids of $W(3,q)$ give $n_b(q+3,4) \geq 2(q^2+1)(q+3)$, both with regular graphs. At diameter 6 an explicit set of $q^2(q^2-1)$ lines of the split Cayley hexagon $H(q)$ ("whiskers") added to Delorme's doubling gives $n_b(q+m,6) \geq 2(q+1)(q^4+q^2+1)+(m-1)\big(2q^4+2q^3+2q+2+m(q^3+q+2)+m^2\big)$, and a regular version. At diameter 7 we join $2r+1$ copies of the incidence graph of $H(q)$ by matchings, delete matching edges at suitable sets and attach new vertices; we prove which sets are allowed. All constructions are given in projective coordinates with the exact neighbourhood of every new vertex. We also show that the graphs of Gómez behind six diameter-8 entries of the table have diameter at least 9 as defined, and that relabelling their copies by automorphisms does not help.

POINTS FOR THE CO-AUTHORS

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- Authors, order, affiliations, e-mail addresses. The byline of [AL20] is "Gabriela Araujo, Nacho López"; use whichever form you prefer here and in the references. 2
- AI disclosure. The constructions, proofs and checks in this draft were produced in a research repository run by Claude Code agents coordinated by D. Özden, and most explicit ingredients (sets, connection sets, hub graphs) were found by computer search. Following the guidance of the 20 October meeting this draft keeps search and AI out of the mathematics and mentions computation only as "computer-checked". Decide how to acknowledge the AI use (most journals now ask for a statement) and whether to cite the repository `https://github.com/dogaozden/degree-diameter`, which holds every graph, builder and check log. 2
- Scope. The repository also holds graphs at other cells that are not covered by the refereed notes this draft is built from: $(16,3)$ 396; $(4,5)$ 162, $(5,5)$ 354, $(8,5)$ 2 048, $(9,5)$ 2 758, $(10,5)$ 4 800, $(11,5)$ 7 020, $(12,5)$ 9 840, $(13,5)$ 11 700, $(14,5)$ 17 568, $(15,5)$ 20 736, $(16,5)$ 33 792 (lifts of generalized quadrangles and other lifts); $(4,7)$ 864, $(5,7)$ 3 120, $(6,7)$ 8 880, $(8,7)$ 43 344; $(3,10)$ 810; and later Haar graphs at $(9,8)$ 546 720, $(5,9)$ 29 756, $(7,10)$ 2 298 240, $(8,10)$ 8 043 840. Decide whether this paper should include them (they would need write-ups and referee reports first), mention them in a remark, or leave them for a second paper. 4

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The referee of the diameter-7 note recalls that this form of the quadric and of Tits' equations is [?, §2.4.13]. We could not consult [VM98] to confirm the place; please check and cite it. 6
Gabriela, Nacho: please check Remark 3.5 against the journal version of [AL20] (we have only seen arXiv v1), and decide whether the remark should stay in the paper. . . 11
Example 4.6: the optimality of 21 comes from an integer-programming solver (HiGHS). Keep the sentence, or drop it as computer-dependent? 13
Proposition 4.10: do you know whether the 2-ovoids $C_a \cup C_{q-a}$ are those of [BLP09], [CCEM08] or [FWX19]? If so, cite and shorten. 16
Corollary 5.15, $q = 2$: the 12 whiskers are a computer-found set (its exactness is also proved in Lean). List them in an appendix, or drop the $q = 2$ case? 23
Theorem 5.18: the completing anti-flags are only shown to exist (as in the Lean proof). If you want every neighbourhood written out, we need an explicit completion (for instance a formula on the deficient elements); this is open work. 23

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1. Introduction

A bipartite graph of maximum degree $\Delta \geq 2$ and diameter $D \geq 2$ has at most

$$M_b(\Delta, D) = 2\big(1 + (\Delta - 1) + \cdots + (\Delta - 1)^{D-1}\big)$$

vertices (the bipartite Moore bound). Let $n_b(\Delta, D)$ be the largest order of such a graph. For $D \geq 3$ and $\Delta \geq 3$ the bound is attained only by the incidence graphs of generalized polygons: projective planes ($D = 3$), generalized quadrangles ($D = 4$) and generalized hexagons ($D = 6$) of order $(\Delta - 1, \Delta - 1)$ [?, ?]. These are known when $\Delta - 1$ is a prime power. For the other cells only lower bounds are known. They are collected in Table 4 of the survey of Miller and Širáň [?] (version of 2013), which the CombinatoricsWiki [?] reproduces; since 2013 the wiki has changed only at $(13, 3)$ and $(15, 3)$, by the work of Araujo-Pardo and López [?]. We call this table "the 2013 table".

This paper gives constructions that improve the 2013 table in the cells listed in Table 1. Most of them are built from generalized polygons in their classical coordinates, by three kinds of operations: copying points and lines (with Delorme's doubling with levels [?] as the basic tool), changing a bounded number of incidences, and joining several copies of an incidence graph by matchings and new vertices. The remaining cells are Haar graphs (bi-Cayley graphs) of explicit groups.

Results.

- *Diameter 3* (Section 3). Araujo-Pardo and López [**?**] added three "hats" to $\mathrm{PG}(2,q)$ and obtained $n_b(q+2,3) \geq 2(q^2+q+5)$. We add a fourth hat (Theorem 3.2): $n_b(q+2,3) \geq 2(q^2+q+6)$ for every prime power $q \geq 4$, attained by a $(q+2)$-regular graph; the construction works in every projective plane of order $q \geq 3$.
- *Diameter 4* (Section 4). Theorem 4.2: in any generalized quadrangle, clones of points with arbitrary weights and copies of lines with "stand-ins" keep diameter 4. With a weighted 2-ovoid of the Ahrens–Szekeres quadrangle $\mathrm{AS}(q)$ made of two elliptic quadrics (Theorem 4.7) this gives $n_b(q+2,4) \geq 2q^2(q+2)$ for odd q; with the Singer 2-ovoids of $W(3,q)$ (Proposition 4.10) it gives $n_b(q+3,4) \geq 2(q^2+1)(q+3)$ for odd q; both with regular graphs. For every prime power $q \leq \Delta - 1$, $n_b(\Delta,4) \geq 2q^2\Delta$ (Corollary 4.13).
- *Diameter 6* (Section 5). A set of $q^2(q^2-1)$ lines of the split Cayley hexagon $H(q)$, given by an explicit rule (Theorem 5.5), can be added at level 0 to Delorme's doubling of $H(q)$ along a Hermitian spread. This gives Theorem 5.8, and deleting repeated lines and completing gives regular graphs (Theorem 5.18).
- *Diameter 7* (Section **??**). The 2013 values at $(7,7)$ and $(9,7)$–$(16,7)$ are the orders $(2r+1)\cdot 2|H(q)|$ of $2r+1$ copies of the incidence graph of $H(q)$ joined by matchings. We delete matching edges at suitable sets of elements and attach new vertices to the freed slots. Theorem **??** reduces the diameter to one condition per pair of copies; Theorems **??** and **??** give two sufficient conditions (segment sets at automorphic transitions; local dualities), the first also necessary. With classical sets this improves all nine cells (Corollaries **??**–**??**); explicit larger sets give the values of Table 1.
- *Diameter 8* (Section **??**). Six entries of the 2013 table at $D = 8$ are the orders of the compound graphs $\Omega\mathrm{H}_q$ of Gómez [**?**], for which the thesis states diameter at most 8 without proof. As defined there, these graphs have diameter at least 9 (Theorems **??** and **??**), and relabelling the copies by automorphisms of the hexagon does not give diameter 8 (Theorems **??** and **??**).
- *Haar graphs* (Section **??**). We state the covering criterion that decides the diameter of a Haar graph from two breadth-first searches, and list Haar graphs that improve two cells at $D = 5$ and 34 cells at $D = 8, 9, 10$. These are single explicit graphs, checked by computer.

Problem 4 of [**?**] asks to generalize the hat construction to regular bipartite graphs of diameters 4 and 6. The hat lemma (Lemma 2.7) says in what sense it generalizes, and Corollaries 4.8 and 4.12 and Theorem 5.18 give regular graphs at $D = 4$ and $D = 6$ that beat the 2013 table.

What is proved and what is computed. Every theorem in Sections 2–**??** has a complete proof here, with one kind of exception, always marked: a statement about one explicit finite object (a set of points, a connection set, a hub graph, a count of route words) is *computer-checked*, and we name the check. "Computer-checked" for a graph means: built from its definition, and found to be bipartite, of the stated maximum degree and of the stated diameter by breadth-first search from every vertex (or from one vertex of every orbit of an automorphism group that was itself checked), by at least two programs written separately. Appendix **??** lists the checks. Many of the statements are also proved in Lean 4 with Mathlib; Appendix **??** says exactly which.

Table 1: The cells improved in this paper. M_b: bipartite Moore bound; 2013: [**?**, Table 4]; "reg.": the largest regular graph we give, when the record graph is not regular. Values at $D = 8, 9, 10$ are in Table **??**.

(Δ, D)	M_b	2013	this paper	reg.	construction
(11,3)	222	190	**192**		fourth hat, $q = 9$ (Thm. 3.2)

(Δ, D)	M_b	2013	this paper	reg.	construction
(13,3)	314	270 [a]	**276**		fourth hat, $q = 11$
(15,3)	422	370 [a]	**376**		fourth hat, $q = 13$
(7,4)	518	346	**359**	350	clones and shadows on $Q(4,5)$ (Ex. 4.6)
(11,4)	2 222	1 734	**1 782**		AS(9), two quadrics (Cor. 4.8)
(13,4)	3 770	3 064	**3 146**		AS(11)
(15,4)	5 910	4 946	**5 070**		AS(13)
(16,4)	7 232	5 134	**5 440**		Singer 2-ovoid of $W(3,13)$ (Cor. 4.12)
(7,5)	3 110	1 134	**1 152**		Haar graph (Sec. **??**)
(9,5)	9 362	2 496	**2 662**		Haar graph
(7,6)	18 662	8 992	**9 592**	9 348	whiskers in $H(5)$ (Thm. 5.8, 5.18)
(11,6)	222 222	142 464	**148 944**	147 500	$H(9)$
(13,6)	542 906	374 452	**388 972**	386 328	$H(11)$
(15,6)	1 158 390	842 048	**870 440**	866 068	$H(13)$
(16,6)	1 627 232	884 062	**940 846**	927 724	$H(13)$, $m = 3$
(7,7)	111 974	23 436	**23 598**		3 levels of $H(5)$, local duality (Sec. **??**)
(9,7)	599 186	117 648	**118 238**		3 levels of $H(7)$
(10,7)	1 195 742	224 694	**225 152**		3 levels of $H(8)$
(11,7)	2 222 222	398 580	**413 180**		3 levels of $H(9)$, duality
(12,7)	3 897 434	664 300	**693 500**		5 levels of $H(9)$, duality
(13,7)	6 514 874	1 062 936	**1 064 472**		3 levels of $H(11)$
(14,7)	10 458 086	1 771 560	**1 789 588**		5 typed levels of $H(11)$
(15,7)	16 217 462	2 480 184	**2 501 491**		7 typed levels of $H(11)$
(16,7)	24 408 482	4 022 340	**4 052 108**		5 typed levels of $H(13)$

[a] The current values of [**?**] are 274 and 376 [**?**].

All graphs in Table 1 except the four Haar graphs at $D = 5$ and $D = 8$–10 are members of families proved here for every prime power q (or every odd q), with explicit ingredients; at $D = 4$ (7,4) and at $D = 7$ the values in the table use, in addition, explicit sets that are computer-checked, and Section **??** also gives values beating the 2013 table in all nine cells from classical ingredients only (Table **??**).

Scope. The repository also holds graphs at other cells that are not covered by the refereed notes this draft is built from: $(16,3)$ 396; $(4,5)$ 162, $(5,5)$ 354, $(8,5)$ 2 048, $(9,5)$ 2 758, $(10,5)$ 4 800, $(11,5)$ 7 020, $(12,5)$ 9 840, $(13,5)$ 11 700, $(14,5)$ 17 568, $(15,5)$ 20 736, $(16,5)$ 33 792 (lifts of generalized quadrangles and other lifts); $(4,7)$ 864, $(5,7)$ 3 120, $(6,7)$ 8 880, $(8,7)$ 43 344; $(3,10)$ 810; and later Haar graphs at $(9,8)$ 546 720, $(5,9)$ 29 756, $(7,10)$ 2 298 240, $(8,10)$ 8 043 840. Decide whether this paper should include them (they would need write-ups and referee reports first), mention them in a remark, or leave them for a second paper.

Organisation. Section 2 fixes notation, coordinates for the classical polygons, and Delorme's doubling with levels, with a proof and the hat lemma. Sections 3–**??** treat diameters 3, 4, 6 and 7, Section **??** the Gómez entries at diameter 8, and Section **??** Haar graphs. Section **??** collects the updated values. Appendix **??** writes out the $(7,4)$ example, Appendix **??** lists what is checked in Lean, and Appendix **??** the computer checks.

2. Preliminaries

2.1. **Incidence structures.** An *incidence structure* $\mathcal{S} = (\mathcal{P}, \mathcal{L}, \mathrm{I})$ has points, lines and a symmetric incidence relation; repeated points and repeated lines (with the same incidences) are allowed.

Its *incidence graph* $\Gamma(\mathcal{S})$ has vertex set $\mathcal{P} \cup \mathcal{L}$ and the incident pairs (*flags*) as edges; it is bipartite with the points on one side. A non-incident point–line pair is an *anti-flag*. Every bipartite graph with a chosen 2-colouring is of this form, so

$$n_b(\Delta, D) = \max\{|\mathcal{P}| + |\mathcal{L}| : \text{ every element has at most } \Delta \text{ neighbours, } \operatorname{diam}\Gamma(\mathcal{S}) \le D\}.$$

Distances $d(x, y)$ are taken in $\Gamma(\mathcal{S})$, and $\operatorname{diam}\mathcal{S}$ means $\operatorname{diam}\Gamma(\mathcal{S})$. Two distinct points on a common line are *collinear*; two distinct lines on a common point are *concurrent*. $\mathbb{F} = \mathrm{GF}(q)$ is the field of order q, and for q odd χ is its quadratic character, with $\chi(0) = 0$.

Lemma 2.1 (diameter in geometric terms)**.** *Let $\mathcal{S}$ have no isolated element and let $D \ge 2$.*

(i) If D is even, $\operatorname{diam}\mathcal{S} \le D$ *if and only if $d(p, L) \le D - 1$ for every anti-flag (p, L).*

(ii) If D is odd, $\operatorname{diam}\mathcal{S} \le D$ *if and only if any two points, and any two lines, are at distance at most $D - 1$.*

In particular, $\operatorname{diam}\mathcal{S} \le 3$ *if and only if any two points are collinear and any two lines are concurrent, and* $\operatorname{diam}\mathcal{S} \le 4$ *if and only if for every anti-flag (p, L) some point of L is collinear with p.*

Proof. (i) Let $p \ne p'$ be points and L a line on p'. If $p \mathrel{\mathrm{I}} L$ then $d(p, p') = 2$; otherwise $d(p, p') \le d(p, L) + 1 \le D$. Lines likewise. (ii) For an anti-flag (p, L) take $p' \mathrel{\mathrm{I}} L$; then $d(p, L) \le d(p, p') + 1 \le D$. The converses hold by parity. □

Lemma 2.2 (adding flags, deleting twins)**.** *(a) Adding flags to an incidence structure keeps its incidence graph bipartite and does not increase any distance. (b) If two lines have the same set of points, deleting one of them does not change any distance between the remaining elements.*

Proof. (a) is clear. (b) A shortest path through the deleted line can use its twin instead; it does not already contain the twin, since then it could be shortened. □

2.2. **Generalized polygons.** A *generalized n-gon* is an incidence structure whose incidence graph has diameter n and girth $2n$; it has *order* (s, t) if every line has $s + 1$ points and every point is on $t + 1$ lines. It is *thick* if $s, t \ge 2$. A generalized 3-gon of order (q, q) is a projective plane of order q, with $q^2 + q + 1$ points and as many lines. A generalized quadrangle (GQ) of order (s, t) has $(s + 1)(st + 1)$ points and $(t + 1)(st + 1)$ lines, and its axiom is: for every anti-flag (p, L) exactly one point of L is collinear with p. A generalized hexagon of order (q, q) has $(q + 1)(q^4 + q^2 + 1)$ points and as many lines. By Feit and Higman [**?**], thick generalized n-gons exist only for $n \in \{2, 3, 4, 6, 8\}$, and a generalized octagon of order (s, s) does not exist; so $M_b(\Delta, D)$ is attained, for $\Delta \ge 3$, only at $D \in \{2, 3, 4, 6\}$.

The projective plane. Points of $\mathrm{PG}(2, q)$ are written $(x : y : z)$ and lines $[u : v : w]$, with $(x : y : z) \mathrm{I} [u : v : w]$ if and only if $ux + vy + wz = 0$.

Quadrangles. $W(3, q)$ is the symplectic quadrangle: the points of $\mathrm{PG}(3, q)$ and the lines of $\mathrm{PG}(3, q)$ that are totally isotropic for a non-degenerate alternating form; it is a GQ of order (q, q) [**?**]. $Q(4, q)$ is the parabolic quadric of $\mathrm{PG}(4, q)$ with the lines it contains; it is a GQ of order (q, q), dual to $W(3, q)$ [**?**, 3.2.1]. The Ahrens–Szekeres quadrangle $\mathrm{AS}(q)$ (q odd) is introduced in coordinates in Section 4.2.

The split Cayley hexagon. Let $V = \mathbb{F}^7$ with coordinates $(X_0, \dots, X_6)$,

$$Q(X) = X_0X_4 + X_1X_5 + X_2X_6 - X_3^2, \qquad b(X, Y) = Q(X + Y) - Q(X) - Q(Y),$$

and let $Q(6, q)$ be the parabolic quadric $\{\langle x\rangle : Q(x) = 0\}$ of $\mathrm{PG}(6, q)$. Points $\langle x\rangle, \langle y\rangle$ of $Q(6, q)$ are *perpendicular*, $x \perp y$, if $b(x, y) = 0$, i.e. if the line xy lies on $Q(6, q)$. The *split Cayley hexagon* $H(q)$ [**?**], [**?**, §2.4] has as points all $N = N(q) = (q^6 - 1)/(q - 1) = (q + 1)(q^4 + q^2 + 1)$ points of

$Q(6,q)$, and as lines the lines of $Q(6,q)$ whose Grassmann coordinates $p_{ij} = x_iy_j - x_jy_i$ satisfy Tits' equations

$$p_{12} = p_{34}, \quad p_{54} = p_{32}, \quad p_{20} = p_{35}, \quad p_{65} = p_{30}, \quad p_{01} = p_{36}, \quad p_{46} = p_{31}. \tag{2.1}$$

For example, the $q+1$ lines through $e_0 = (1,0,\dots,0)$ are $\langle e_0, \lambda e_5 + \mu e_6\rangle$, $(\lambda:\mu) \in \mathrm{PG}(1,q)$; they lie in the totally singular plane $\langle e_0, e_5, e_6\rangle$. The same equations define $H(q)$ for q even, with the quadric written $X_0X_4 + X_1X_5 + X_2X_6 + X_3^2$ (Section 5.5).

The referee of the diameter-7 note recalls that this form of the quadric and of Tits' equations is [**?**, §2.4.13]. We could not consult [VM98] to confirm the place; please check and cite it.

Fact 2.3 ([**?**], [**?**, §2.4])**.** $H(q)$ is a generalized hexagon of order (q,q). The lines of $H(q)$ through a point x are the lines through x of a totally singular plane $\pi_x \subset x^\perp$. The collineation group contains $G_2(q)$; for $A \in \mathrm{SL}(3,q)$ the map $(v,a,w) \mapsto (Av, a, A^{-T}w)$, with $x = (v,a,w)$, $v = (x_0,x_1,x_2)$, $a = x_3$, $w = (x_4,x_5,x_6)$, is a collineation. $H(q)$ has a duality (an incidence-preserving bijection exchanging points and lines) if and only if q is a power of 3.

Let $I = I(q)$ be the incidence graph of $H(q)$, with $2N$ vertices, and $\Gamma_i(x) = \{y : d(x,y) = i\}$. Two elements are *opposite* if they are at distance 6, and $\mathrm{opp}(x) = \Gamma_6(x)$. The following facts are standard; we include the short proofs because Sections **??** and **??** use them repeatedly.

Lemma 2.4. *Let x, y be vertices of I.*

(a) If $d(x,y) \le 5$ there is a unique geodesic from x to y. If $1 \le d(x,y) = k \le 5$, exactly one neighbour of y is at distance $k-1$ from x and the other q at distance $k+1$; if $k = 6$ all $q+1$ neighbours of y are at distance 5. Hence $|\Gamma_i(x)| = (q+1)q^{i-1}$ for $1 \le i \le 5$ and $|\Gamma_6(x)| = q^5$.

(b) (covering) *There is $z \in \Gamma_2(x)$ with $d(z,y) \le 3$ if x, y have different types, and with $d(z,y) \le 4$ if they have the same type.*

(c) (orthogonality) *For points x, y: $d(x,y) \le 4$ if and only if $x \perp y$. So $\mathrm{opp}(x) \cap \mathcal{P} = \mathcal{P} \setminus x^\perp$.*

Proof. (a) Two geodesics would close a cycle of length at most $10 < 12$. Neighbours of y are at distance $k \pm 1$ from x; the predecessor on the geodesic is at $k-1$, and a second one would give a second geodesic; at $k = 6$ the diameter forces 5. The counts follow by induction. (b) If $2 \le d(x,y) \le 6$ take the vertex at distance 2 from x on a geodesic to y; if $d(x,y) = 1$ take $z \in \Gamma_1(y) \setminus \{x\}$; if $x = y$ any $z \in \Gamma_2(x)$. The largest value of $d(z,y)$ is $6-2 = 4$ (same type) or $5-2 = 3$ (different types). (c) If $d(x,y) \le 4$ there is a point z with x, y on lines of $H(q)$ through z ($z = x$ if $d(x,y) \le 2$); these lines lie in π_z, so $b(x,y) = 0$. Conversely $x^\perp \cap Q(6,q)$ is a cone with vertex x over a $Q(4,q)$ and has $1 + q(q^3+q^2+q+1)$ points, which by (a) is the number $1 + (q+1)q + (q+1)q^3$ of points at distance at most 4 from x. □

2.3. **Doubling with levels, and the hat lemma.** A *duplicate* of a vertex (a new vertex with the same neighbours) does not increase the diameter (see [**?**], which we have not seen, and [**?**, §7]). Delorme [**?**, §7] refined this into a doubling with levels, stated there without proof, and used it for the 2013 values at $D = 4$ and $D = 6$ [**?**, §8(a),(b)]. We state it for incidence structures and give a proof.

Let $\mathcal{A} = (\mathcal{A}_0 \supseteq \mathcal{A}_1 \supseteq \cdots \supseteq \mathcal{A}_r \supseteq \mathcal{A}_{r+1} = \emptyset)$ be a filtration of a set $\mathcal{A}_0$ of elements (points and lines may be mixed). The *level* of x is $\lambda(x) = \max\{k : x \in \mathcal{A}_k\}$, and $\lambda(x) = -1$ if $x \notin \mathcal{A}_0$. Fix $m \ge 2$. Two words over $[m] = \{1,\dots,m\}$ are *comparable* if one is a prefix of the other.

Definition 2.5 (Delorme)**.** The m-fold doubling $\mathcal{S}_{\mathcal{A}}^{(m)}$ has the *versions* x^w, for x an element of $\mathcal{S}$ and $w \in [m]^{\lambda(x)+1}$ (x^w is a point if x is a point and a line if x is a line), and

$$x^w \mathrel{\mathrm{I}} y^u \iff x \mathrel{\mathrm{I}} y \text{ and } w, u \text{ are comparable.}$$

So an element of level -1 has one version (identified with x), adjacent to every version of each of its neighbours; and a version x^w of level $k \geq 0$ is adjacent, for each neighbour y of x, to the single version $y^{w_1 \cdots w_{j+1}}$ if $\lambda(y) = j \leq k$, and to the m^{j-k} versions y^{wu} ($|u| = j - k$) if $\lambda(y) = j > k$. This is the exact neighbourhood of every version. The map $x^w \mapsto x$ preserves incidence, so $d(x^w, y^u) \geq d(x, y)$.

The conditions on $\mathcal{A}$ are (with $D \geq 2$):

(L0) every element outside $\mathcal{A}_0$ is incident with at most one element of $\mathcal{A}_0$;
(L1) for every $i \geq 0$: every element incident with an element of $\mathcal{A}_{i+1}$ lies in $\mathcal{A}_i$, and every element of $\mathcal{A}_i \setminus \mathcal{A}_{i+1}$ is incident with at most one element of $\mathcal{A}_{i+1}$;
(C_D) for any two elements x, y (possibly $x = y$) in the same connected component of the substructure induced on $\mathcal{A}_0$, there is $z \notin \mathcal{A}_0$ with $d(x, z) + d(z, y) \leq D$.

Theorem 2.6 (Delorme [**?**, §7], (a)–(c))**.** *Let $\mathcal{S}$ have diameter at most $D \geq 2$ and let $m \geq 2$.*

(a) If (C_D) holds, $\mathcal{S}_{\mathcal{A}}^{(m)}$ has diameter at most D.
(b) If (L0) and (L1) hold, every version satisfies $\deg x^w \leq \deg x + m - 1$, with equality only if x is incident with an element of level $\lambda(x) + 1$.
(c) $\mathcal{S}_{\mathcal{A}}^{(m)}$ has $|\mathcal{P}| + (m-1)\sum_{k\geq 0} m^k |\mathcal{A}_k \cap \mathcal{P}|$ points and $|\mathcal{L}| + (m-1)\sum_{k\geq 0} m^k |\mathcal{A}_k \cap \mathcal{L}|$ lines.
(d) Conversely, if $\mathcal{S}_{\mathcal{A}}^{(m)}$ has diameter at most D, the conclusion of (C_D) holds for every pair $x, y \in \mathcal{A}_0$.

Delorme states (a)–(c) without proof ("je ne donne pas la preuve"); (d) is not in [**?**].

Proof. (c) An element of level k has $m^{k+1} = 1 + (m-1)(1 + m + \cdots + m^k)$ versions.

(b) Let $\lambda(x) = k$. By the description after Definition 2.5, only neighbours of level $> k$ contribute more than one version. If $k = -1$, (L0) gives at most one neighbour in $\mathcal{A}_0$; it has level exactly 0 (a neighbour in $\mathcal{A}_1$ would put x in $\mathcal{A}_0$ by (L1)), and contributes m versions. If $k \geq 0$, (L1) with $i = k$ gives at most one neighbour in $\mathcal{A}_{k+1}$, and (L1) with $i = k + 1$ gives none in $\mathcal{A}_{k+2}$; that neighbour contributes m versions. So $\deg x^w \leq \deg x - 1 + m$.

(a) For an infinite word ω the *sheet* Σ_ω is the set of versions x^w with w a prefix of ω. It contains one version of each element, and $x^w \mapsto x$ is an isomorphism from the substructure on Σ_ω onto $\mathcal{S}$. Two versions with comparable words lie in a common sheet, hence are at distance at most D. Let x^w, y^u have incomparable words; both are non-empty, so $x, y \in \mathcal{A}_0$. If x, y are in different components of the substructure on $\mathcal{A}_0$, a shortest path $x = z_0, \ldots, z_k = y$ ($k \leq D$) passes through some $z_i \notin \mathcal{A}_0$, whose unique version lies in every sheet. Lift $z_0, \ldots, z_i$ into a sheet through x^w and $z_i, \ldots, z_k$ into a sheet through y^u: this is a path of length k. If x, y are in the same component (including $x = y$), (C_D) gives $z \notin \mathcal{A}_0$ with $d(x, z) + d(z, y) \leq D$, and the same lifting gives a path of length at most D.

(d) Take versions x^w, y^u with w starting with the letter 1 and u with the letter 2, and a path of length at most D between them. Consecutive versions on it have comparable words, and comparable non-empty words start with the same letter; so some version on the path has the empty word, that is, it is an element $z \notin \mathcal{A}_0$. Projecting the path gives a walk $x \to z \to y$ of length at most D. □

Lemma 2.7 (hat lemma)**.** *Let $\mathcal{S}$ be a generalized n-gon of order (s, t), $s, t \geq 1$, and $\mathcal{A}_0$ a set of elements. Then (C_n) holds if and only if every connected component of the substructure induced on $\mathcal{A}_0$ is a tree of diameter at most $n - 2$.*

Proof. Suppose a component B contains a path $x = z_0, \ldots, z_{n-1} = y$ of length $n - 1$ (it does if B has diameter at least $n - 1$, and if B contains a cycle, since cycles have length at least $2n$). By the girth this is the only path of length at most $n - 1$ from x to y, so $d(x, y) = n - 1$. If

$d(x,z)+d(z,y)\le n$ then by parity $d(x,z)+d(z,y)=n-1$, so z lies on a shortest x–y path, i.e. $z=z_i\in\mathcal{A}_0$. So (C_n) fails.

Conversely let B be a tree of diameter at most $n-2$, $x,y\in B$, and P the path from x to y in B, of length d. The incidence graph is connected and has cycles, so some element of B has a neighbour outside $\mathcal{A}_0$. Let k be the least distance in B from P to such an element u, and z a neighbour of u outside $\mathcal{A}_0$. Then $d(x,z)+d(z,y)\le d+2k+2$. If $k=0$ this is at most n. If $k\ge 1$, every element of B at distance less than k from P has all its neighbours in B, and every element has at least two neighbours. So we can walk k steps away from P starting at x, and k steps away from P starting at y (along two different neighbours if $x=y$); in the tree B this gives a path of length $d+2k$. Hence $d+2k\le n-2$ and $d(x,z)+d(z,y)\le n$. □

Corollary 2.8. *In a generalized n-gon of order (s,t), $s,t\ge 1$, for any filtration and $m\ge 2$, $\mathcal{S}_{\mathcal{A}}^{(m)}$ has diameter at most n if and only if every component of the substructure on $\mathcal{A}_0$ is a tree of diameter at most $n-2$. For $n=3$ the allowed components are single elements and flags; for $n=4$ also* stars *(a point with some lines on it, or a line with some of its points); for $n=6$ trees of diameter at most 4.*

Proposition 2.9 (diameter 3 needs patches)**.** *In a projective plane, if $\mathcal{A}$ satisfies (L0), (L1) and (C_3), then $\mathcal{A}_0$ has at most one point and at most one line, and $\mathcal{A}_1=\emptyset$.*

Proof. If $x\ne y$ are points of $\mathcal{A}_0$, the line xy is either outside $\mathcal{A}_0$ and incident with two elements of $\mathcal{A}_0$ (against (L0)), or in $\mathcal{A}_0$, and then x,xy,y is a path of length $2>n-2$ in one component (against Lemma 2.7). Lines likewise. An element of $\mathcal{A}_1$ would have its $q+1\ge 3$ neighbours in $\mathcal{A}_0$ by (L1). □

So pure doubling gains at most $2(m-1)$ elements in a projective plane; the hats of Section 3 double flags and then change a bounded number of incidences.

3. Diameter 3: the fourth hat

Araujo-Pardo and López [**?**] add three *hats* to $\mathrm{PG}(2,q)$ and obtain $n_b(q+2,3)\ge 2(q^2+q+5)$. We add a fourth. By Proposition 2.9, every hat must change some incidences of the plane; the fourth hat costs one rotation of three incidences.

3.1. **Frame and notation.** Let $\ell=[1:0:0]$ (the line $x=0$) and $A=(1:0:0)\notin\ell$. The points of ℓ and the lines on A are

$$X_t=(0:t:1),\quad P=(0:1:0);\qquad \lambda_t=[0:1:-t],\quad \lambda_\infty=[0:0:1]\qquad (t\in\mathbb{F}).$$

Every point is listed once among A, X_t ($t\in\mathbb{F}$), P, $(x:y:1)$ ($x\ne 0$), $(x:1:0)$ ($x\ne 0$); every line once among ℓ, λ_t, λ_∞, $[u:1:w]$ ($u\ne 0$), $[1:0:w]$ ($w\ne 0$). We use these facts (one-line computations):

(F1) $\lambda_t\cap\ell=X_t$, $\lambda_\infty\cap\ell=P$; the points of λ_t are A and $(x:t:1)$, $x\in\mathbb{F}$ (X_t at $x=0$); those of λ_∞ are A and $(x:1:0)$, $x\in\mathbb{F}$.
(F2) The lines on X_t are ℓ and $[w:1:-t]$, $w\in\mathbb{F}$ (λ_t at $w=0$); the lines on P are ℓ, λ_∞ and $[1:0:w]$, $w\ne 0$.
(F3) If $u=(x:y:z)$ with $x\ne 0$, then u and X_s lie on $[(sz-y)/x:1:-s]\ne\ell$.
(F4) A line $[u:v:w]$ with $u\ne 0$ meets λ_s in $(-(vs+w)/u:s:1)\ne A$; $[u:1:w]$ meets ℓ in X_{-w}.

Let $q\ge 4$, let $a,c\in\mathbb{F}\setminus\{0,1\}$ be distinct, and put

$$S=\{0,1,a,c\},\qquad T=\{0,1,c\},\qquad \sigma=(0\ c\ 1):\ 0\mapsto c\mapsto 1\mapsto 0.$$

3.2. **The construction.**

Definition 3.1. $\mathcal{S}(q;a,c)$ has the points of $\mathrm{PG}(2,q)$ and five new points Q, H_0, H_1, H_a, H_c, the lines of $\mathrm{PG}(2,q)$ and five new lines M, K_0, K_1, K_a, K_c, and exactly these incidences:

(C1) every incidence of $\mathrm{PG}(2,q)$ except $X_s \mathrel{\mathrm{I}} \lambda_s$ ($s \in S$), and the *rotation* $X_0 \mathrel{\mathrm{I}} \lambda_c$, $X_c \mathrel{\mathrm{I}} \lambda_1$, $X_1 \mathrel{\mathrm{I}} \lambda_0$ (that is, $X_t \mathrel{\mathrm{I}} \lambda_{\sigma(t)}$ for $t \in T$);
(C2) H_s lies on $[w:1:-s]$ for every $w \in \mathbb{F}$ ($s \in S$): the lines on X_s other than ℓ;
(C3) K_s contains $(x:s:1)$ for every $x \in \mathbb{F}$ ($s \in S$): the points of λ_s other than A;
(C4) M contains P, X_t ($t \in \mathbb{F} \setminus T$), and H_0, H_1, H_a, H_c;
(C5) Q lies on λ_∞, λ_t ($t \in \mathbb{F} \setminus T$), and K_0, K_1, K_a, K_c;
(C6) the *links* $A \mathrel{\mathrm{I}} K_a$, $H_a \mathrel{\mathrm{I}} \ell$, $H_0 \mathrel{\mathrm{I}} K_c$, $H_c \mathrel{\mathrm{I}} K_1$, $H_1 \mathrel{\mathrm{I}} K_0$ (that is, $H_t \mathrel{\mathrm{I}} K_{\sigma(t)}$ for $t \in T$).

$G(q;a,c)$ is its incidence graph. $R(q;a,c)$ is $G(q;a,c)$ together with the *filler* edges

$$(x:y:1) \sim [x:1:-y] \quad (x \neq 0,\ y \notin S), \qquad (x:1:0) \sim [1:0:-1/x] \quad (x \neq 0).$$

The hat pair (H_s, K_s) is a copy of the flag (X_s, λ_s) with the flag "crossed", and (Q, M) is a copy of the anti-flag (A, ℓ).

Theorem 3.2. *For every prime power $q \geq 4$ and distinct $a, c \in \mathbb{F} \setminus \{0,1\}$, the graphs $G(q;a,c)$ and $R(q;a,c)$ are bipartite, have $2(q^2+q+6)$ vertices, maximum degree $q+2$ and diameter 3, and $R(q;a,c)$ is $(q+2)$-regular. Hence*

$$n_b(q+2,3) \geq 2(q^2+q+6),$$

attained by a regular graph.

The construction and the proof use only the axioms of a projective plane together with an anti-flag (A,ℓ) and four points X_s of ℓ, so a coordinate-free version holds in every projective plane of order $q \geq 3$ (in coordinates the four points are labelled by $0, 1, a, c$, which needs $q \geq 4$). It also has a regular completion there. At $q = 3$ the four points X_s are all of ℓ, there is no deficient element, and the structure is already 5-regular. For $q \geq 4$ put $N_0 = (q-1)(q-3)$, the number of points of degree $q+1$ and of lines of degree $q+1$ (see the table below). A deficient point p lies on exactly $q-4$ deficient lines: the lines on p other than Ap meet ℓ in the q points other than $Ap \cap \ell$, four of which are the X_s. Dually, a deficient line has exactly $q-4$ deficient points. So the bipartite graph of non-incident pairs between deficient points and deficient lines is regular of degree $N_0 - (q-4) = q^2 - 5q + 7 \geq 1$, and a regular bipartite graph of positive degree has a perfect matching (Kőnig); adding it as flags completes the structure. In coordinates, the filler of Definition 3.1 is such a matching.

3.3. **Every neighbourhood.** Table 2 lists the neighbourhood in $G(q;a,c)$ of every vertex, new or old. Here $\sigma^{-1}(0) = 1$, $\sigma^{-1}(1) = c$, $\sigma^{-1}(c) = 0$.

Table 2: The neighbourhood of every vertex of $G(q;a,c)$.

vertex	neighbours	degree
Q	λ_∞, λ_t ($t \notin T$), K_0, K_1, K_a, K_c	$q+2$
H_a	$[w:1:-a]$ ($w \in \mathbb{F}$), M, ℓ	$q+2$
H_t, $t \in T$	$[w:1:-t]$ ($w \in \mathbb{F}$), M, $K_{\sigma(t)}$	$q+2$
M	P, X_t ($t \notin T$), H_0, H_1, H_a, H_c	$q+2$
K_a	$(x:a:1)$ ($x \in \mathbb{F}$), Q, A	$q+2$
K_t, $t \in T$	$(x:t:1)$ ($x \in \mathbb{F}$), Q, $H_{\sigma^{-1}(t)}$	$q+2$
A	λ_t ($t \in \mathbb{F}$), λ_∞, K_a	$q+2$
P	ℓ, λ_∞, $[1:0:w]$ ($w \neq 0$), M	$q+2$

vertex	neighbours	degree
X_t, $t \notin S$	ℓ, $[w:1:-t]$ $(w \in \mathbb{F})$, M	$q+2$
X_a	ℓ, $[w:1:-a]$ $(w \neq 0)$, M, K_a	$q+2$
X_t, $t \in T$	ℓ, $[w:1:-t]$ $(w \neq 0)$, $\lambda_{\sigma(t)}$, K_t	$q+2$
$(x:y:1)$, $x \neq 0$	$[u:1:-(ux+y)]$ $(u \in \mathbb{F})$, $[1:0:-x]$, and K_y if $y \in S$	$q+2$ if $y \in S$, else $q+1$
$(x:1:0)$, $x \neq 0$	λ_∞, $[1:-x:w]$ $(w \in \mathbb{F})$	$q+1$
ℓ	P, X_t $(t \in \mathbb{F})$, H_a	$q+2$
λ_∞	A, $(x:1:0)$ $(x \in \mathbb{F})$, Q	$q+2$
λ_t, $t \notin S$	A, $(x:t:1)$ $(x \in \mathbb{F})$, Q	$q+2$
λ_a	A, $(x:a:1)$ $(x \neq 0)$, Q, H_a	$q+2$
λ_t, $t \in T$	A, $(x:t:1)$ $(x \neq 0)$, $X_{\sigma^{-1}(t)}$, H_t	$q+2$
$[u:1:w]$, $u \neq 0$	$(x:-ux-w:1)$ $(x \in \mathbb{F})$, $(-1/u:1:0)$, and H_{-w} if $-w \in S$	$q+2$ if $-w \in S$, else $q+1$
$[1:0:w]$, $w \neq 0$	$(-w:y:1)$ $(y \in \mathbb{F})$, P	$q+1$

The vertices of degree $q+1$ are the points $(x:y:1)$ $(x \neq 0,\ y \notin S)$ and $(x:1:0)$ $(x \neq 0)$, and the lines $[u:1:w]$ $(u \neq 0,\ -w \notin S)$ and $[1:0:w]$ $(w \neq 0)$: $(q-1)(q-3)$ of each kind. In $R(q;a,c)$ each of them gains exactly one neighbour, its filler partner.

3.4. **Proof of Theorem 3.2.** *Degrees.* Each row of Table 2 is (F1) or (F2) corrected by (C1)–(C6), and within a row the listed neighbours are distinct. For instance: A keeps its $q+1$ lines and gains only K_a (A is not of the form $(x:s:1)$); X_a loses λ_a and gains M (as $a \notin T$) and K_a; X_t $(t \in T)$ loses λ_t, gains $\lambda_{\sigma(t)}$ and K_t, and is not on M; a point $(x:y:1)$ with $x \neq 0$ is not on ℓ, so it loses nothing, and it gains K_s if and only if $y = s \in S$; Q is on $1+(q-3)+4 = q+2$ lines; H_s is on q lines by (C2), on M, and on exactly one link line. The line rows are the dual count: ℓ gains only H_a; λ_a loses X_a and gains Q, H_a; λ_t $(t \in T)$ loses X_t and gains $X_{\sigma^{-1}(t)}$ and H_t; $[u:1:w]$ $(u \neq 0)$ gains H_s if and only if $s = -w \in S$; M has $1+(q-3)+4$ points; K_s has q points, Q, and one link point.

The filler map $\delta : (x:y:z) \mapsto [x:z:-y]$ is a correlation of $\mathrm{PG}(2,q)$ (since $ux+wz+(-v)(-y) = ux+vy+wz$), a point lies on its image if and only if $x^2 = 0$, i.e. if and only if it is on ℓ, and δ maps $(x:y:1) \mapsto [x:1:-y]$ and $(x:1:0) \mapsto [x:0:-1] = [1:0:-1/x]$. So the filler is a bijection from the deficient points to the deficient lines, and its pairs are anti-flags. Hence $G(q;a,c)$ has maximum degree $q+2$ and $R(q;a,c)$ is $(q+2)$-regular.

Reduction. By Lemma 2.1(ii) it suffices to show that any two points are collinear and any two lines concurrent. In the next two lemmas "incident" means incident in $\mathcal{S}(q;a,c)$. The only incidences of $\mathrm{PG}(2,q)$ that are missing are X_s I λ_s, $s \in S$; so an incidence u I L of the plane survives whenever $u \notin \ell$ or L is not on A.

Lemma 3.3. *Any two points $u \neq v$ of $\mathcal{S}(q;a,c)$ are collinear.*

Proof. (P1) Both in $\mathrm{PG}(2,q)$; let $L = uv$. If both incidences survive, use L. Otherwise, say $u = X_s$ and $L = \lambda_s$ with $s \in S$, and $v = A$ or $v = (x:s:1)$ with $x \neq 0$. If $v = (x:s:1)$: use K_s. If $v = A$ and $s = a$: K_a (by (C3), (C6)). If $v = A$ and $s \in T$: $\lambda_{\sigma(s)}$ (rotation).

(P2) Q and u. If $u \in \{A, P\}$ or $u = (x:1:0)$: λ_∞. If $u = X_t$, $t \notin S$: λ_t. If $u = X_s$, $s \in S$: K_s. If $u = (x:y:1)$, $x \neq 0$: λ_y if $y \notin T$, and K_y if $y \in T$.

(P3) H_s and u. If $u \notin \ell$ (including $u = A$): the line of (F3), which contains H_s by (C2). If $u = X_s$: $[1:1:-s]$. If $u = P$, or $u = X_t$ with $t \notin T$, $t \neq s$: M. If $u = X_t$ with $t \in T \setminus \{s\}$ and $s = a$: ℓ. If $u = X_t$ with $t \in T \setminus \{s\}$ and $s \in T$: as σ is a 3-cycle, $t = \sigma(s)$ or $t = \sigma^{-1}(s)$; use

$K_{\sigma(s)}$ (it contains $X_{\sigma(s)}$ and H_s) or λ_s (it contains $X_{\sigma^{-1}(s)}$ by the rotation, and H_s by (C2) with $w = 0$).

(P4) Two new points. $H_s, H_{s'}$: M. Q, H_a: λ_a. Q, H_t $(t \in T)$: $K_{\sigma(t)}$. □

Lemma 3.4. *Any two lines $L \neq N$ of $\mathcal{S}(q; a, c)$ are concurrent.*

Proof. (N1) Both in $\mathrm{PG}(2, q)$; let $p = L \cap N$. If both incidences survive, use p. Otherwise, say $p = X_s$ and $L = \lambda_s$, and $N = \ell$ or $N = [w : 1 : -s]$ with $w \neq 0$. If $N = [w : 1 : -s]$: H_s. If $N = \ell$ and $s = a$: H_a. If $N = \ell$ and $s \in T$: $X_{\sigma^{-1}(s)}$ (rotation).

(N2) M and L. If L is on P ($L = \ell$, λ_∞ or $[1 : 0 : w]$): P. If $L = \lambda_t$, $t \notin S$: X_t. If $L = \lambda_s$, $s \in S$: H_s. If $L = [u : 1 : w]$, $u \neq 0$: it meets ℓ in X_r, $r = -w$ (F4); use X_r if $r \notin T$, and H_r if $r \in T$ (H_r is on $[u : 1 : -r]$ and on M).

(N3) K_s and L. If L is not on A ($L = \ell$ or $L = [u : v : w]$ with $u \neq 0$): $L \cap \lambda_s = (x : s : 1)$ (F1, F4), which is on K_s, and its incidence with L survives. If $L = \lambda_s$: $(1 : s : 1)$. If $L = \lambda_\infty$ or $L = \lambda_t$, $t \notin T$: Q. If $L = \lambda_t$ with $t \in T \setminus \{s\}$ and $s = a$: A. If $L = \lambda_t$ with $t \in T \setminus \{s\}$ and $s \in T$: if $t = \sigma(s)$, use X_s (on $\lambda_{\sigma(s)}$ and K_s); if $t = \sigma^{-1}(s)$, use H_t (on λ_t and on $K_{\sigma(t)} = K_s$).

(N4) Two new lines. $K_s, K_{s'}$: Q. M, K_a: X_a. M, K_t $(t \in T)$: $H_{\sigma^{-1}(t)}$. □

Proof of Theorem 3.2. By Lemmas 3.3 and 3.4 and Lemma 2.1(ii), $G(q; a, c)$ has diameter at most 3, and adding the filler does not increase it (Lemma 2.2). The graphs are bipartite and not complete bipartite ($A \notin \ell$), so the diameter is exactly 3. The order is $2(q^2 + q + 1) + 10$. □

The proof uses only that $\mathbb{F}$ is a field and that $0, 1, a, c$ are distinct. The correlation δ, extended by $H_s \leftrightarrow K_s$ and $Q \leftrightarrow M$, maps $\mathcal{S}(q; a, c)$ built with σ onto the dual of the structure built with σ^{-1}, so Lemma 3.4 is also Lemma 3.3 for σ^{-1}; we wrote both out so that every case can be checked directly. Both the rotation and the link $H_1 \mathrel{\mathrm{I}} K_0$ are needed: without either, the diameter is 4 in every case we checked by computer (every prime power $q \leq 13$ and every admissible (a, c)).

Remark 3.5 (the three hats of [**?**] in these coordinates). The structure of [**?**] has hats $\{0, 1, a\}$, no rotation, only the flag (X_a, λ_a) removed, links $A \mathrel{\mathrm{I}} K_a$, $H_a \mathrel{\mathrm{I}} \ell$, $H_0 \mathrel{\mathrm{I}} K_1$, $H_1 \mathrel{\mathrm{I}} K_0$; M contains P and X_t $(t \neq 0, 1)$, and Q lies on λ_∞ and λ_t $(t \neq 0, 1)$. The dictionary with the affine coordinates of [**?**] is $(x, y) = (x : y : 1)$, $[m, b] = [m : -1 : b]$, $L_c = [1 : 0 : -c]$, $L = [0 : 0 : 1]$, $P_m = (1 : m : 0)$; so $L_0 = \ell$, $P_0 = A$, $(0, s) = X_s$, $[0, s] = \lambda_s$ and $L = \lambda_\infty$. In the completing matching of [**?**, arXiv v1, p. 8] the index set of $P_i \sim L_i$ should be $\mathbb{F}^*$; with $\mathbb{F} \setminus \{0, 1, a\}$ the four vertices P_1, P_a, L_1, L_a keep degree $q + 1$. Neither the order nor the diameter is affected.

Gabriela, Nacho: please check Remark 3.5 against the journal version of [AL20] (we have only seen arXiv v1), and decide whether the remark should stay in the paper.

3.5. **Cells.** For $q = 9, 11, 13$, Theorem 3.2 gives $(11, 3)$: 192 (2013 table: 190), $(13, 3)$: 276 (2013: 270; current value 274 [**?**]) and $(15, 3)$: 376 (2013: 370; current value 374 [**?**]), all with regular graphs. (The 2013 value 190 at $(11, 3)$ is due to Feria-Purón, Miller and Pineda-Villavicencio [**?**] and, independently, to [**?**].) At $q = 5$ it gives $72 < 80 = n_b(7, 3)$, a value known to be exact [**?**].

4. Diameter 4: clones and shadow lines in generalized quadrangles

4.1. **Clones and shadow lines.** Copies of points and lines of a GQ go back to Delorme [**?**, §§7–8]: his doubling may copy any induced substructure, points and lines together, and his $D = 4$ construction (Example 4.4) doubles a spread. Araujo-Pardo, Dalfó, Fiol and López [**?**, Sec. 4] double every line of a GQ to get biregular graphs. The construction below is not a doubling: a clone is joined to every old line through its point, so no tree condition on the copied set is needed, and it allows weights larger than one, repeated lines, and "stand-ins". (A level-0 line all of whose points are at level -1 is a shadow line with trivial stand-ins.)

Definition 4.1 (clones and shadow lines)**.** Let $\mathcal{S} = (\mathcal{P}, \mathcal{L})$ be a GQ of order (s,t), let $\omega : \mathcal{P} \to \mathbb{Z}_{\geq 0}$ be a *weight*, and $\mathcal{M}$ a finite multiset of lines. For each entry μ of $\mathcal{M}$, with underlying line L_μ, choose for every point p I L_μ a *stand-in* $\theta_\mu(p) \in \{p, p^1, \ldots, p^{\omega(p)}\}$. The incidence structure $\mathcal{S}[\omega, \mathcal{M}, \theta]$ has

- points: $\mathcal{P}$ and the *clones* $p^1, \ldots, p^{\omega(p)}$ of each $p \in \mathcal{P}$;
- lines: $\mathcal{L}$ and one *shadow line* L^μ for each entry μ of $\mathcal{M}$;
- incidences: those of $\mathcal{S}$; p^i I L whenever p I $L \in \mathcal{L}$; and $\theta_\mu(p)$ I L^μ for p I L_μ.

Table 3. Neighbourhoods in $\mathcal{S}[\omega, \mathcal{M}, \theta]$.

vertex	neighbours	degree
clone p^i	the $t+1$ old lines on p; the shadow lines L^μ with $\theta_\mu(p) = p^i$	$t + 1 + \#\{\mu : \theta_\mu(p) = p^i\}$
shadow line L^μ	$\theta_\mu(p)$ for the $s+1$ points p of L_μ	$s+1$
old point p	its t+1 lines; the shadow lines L^μ with $\theta_\mu(p) = p$	$t + 1 + \#\{\mu : \theta_\mu(p) = p\}$
old line L	its $s+1$ points and all their clones	$s + 1 + \sum_{p \mathrm{I} L} \omega(p)$

When all stand-ins are old points ($\theta = \mathrm{id}$), a shadow line is adjacent exactly to the old points of L_μ, and clones are adjacent only to old lines.

Theorem 4.2 (clones and shadow lines)**.** *Let $\mathcal{S}$ be a GQ of order (s,t), $s,t \geq 1$, and $\mathcal{S}' = \mathcal{S}[\omega, \mathcal{M}, \theta]$.*

(a) $\mathcal{S}'$ has diameter 4.

(b) $\mathcal{S}'$ has maximum degree at most Δ if and only if $\Delta \geq \max(s,t) + 1$, $\sum_{p \mathrm{I} L} \omega(p) \leq \Delta - s - 1$ for every line L of $\mathcal{S}$, and every point of $\mathcal{S}'$ (old point or clone) is the stand-in for at most $\Delta - t - 1$ entries of $\mathcal{M}$.

(c) $\mathcal{S}'$ has $(s+1)(st+1) + \sum_p \omega(p)$ points and $(t+1)(st+1) + |\mathcal{M}|$ lines.

Proof. (b) and (c) are read off Table 3. (a) Write π for the map sending a clone or a shadow line to its underlying point or line. Every point x of $\mathcal{S}'$ is incident with every old line through $\pi(x)$. We use Lemma 2.1(i); no element is isolated. Let (x, Y) be an anti-flag of $\mathcal{S}'$.

- Y old. Then $\pi(x) \notin Y$. Let r be the point of Y collinear with $\pi(x)$, on the line N. Then x I N I r I Y.
- $Y = L^\mu$ and $\pi(x) \notin L_\mu$. Let r I L_μ be collinear with $\pi(x)$, on the line N. Then x I N I $\theta_\mu(r)$ I L^μ.
- $Y = L^\mu$ and $\pi(x)$ I L_μ. Pick r I L_μ, $r \neq \pi(x)$ ($s \geq 1$). Then x I L_μ I $\theta_\mu(r)$ I L^μ.

So $d(x, Y) \leq 3$ for every anti-flag, and the diameter is at most 4. Two non-collinear old points exist (a point is collinear with $s(t+1) < (s+1)(st+1) - 1$ others) and stay non-collinear: they share no old line, a shadow line L^μ contains only points over points of L_μ, and an old point lies over itself only. So the diameter is 4. □

Corollary 4.3 (exact weights, regular completion)**.** *Let $\Delta \geq \max(s,t) + 1$, let ω be a weighted $(\Delta - s - 1)$-ovoid (that is, $\sum_{p \mathrm{I} L} \omega(p) = \Delta - s - 1$ for every line L) and $\mathcal{M}$ a weighted $(\Delta - t - 1)$-cover (every point is on exactly $\Delta - t - 1$ entries of $\mathcal{M}$). Then $\mathcal{S}[\omega, \mathcal{M}, \mathrm{id}]$ has diameter 4, maximum degree Δ and $2\Delta(st+1)$ elements. It becomes Δ-regular, with the same order and diameter 4, by the following explicit flags: number the clones $c_0, c_1, \ldots$ and the shadow lines $y_0, \ldots, y_{b-1}$ ($b = |\mathcal{M}|$), put $k = \Delta - t - 1$, and join c_i to y_j for $j \equiv ik, ik+1, \ldots, ik+k-1 \pmod b$. (If $b = 0$ or $\sum_p \omega(p) = 0$, the structure is already Δ-regular.)*

Proof. Counting flags gives $\sum_p \omega(p) = (\Delta-s-1)(st+1)$ and $|\mathcal{M}| = (\Delta-t-1)(st+1)$; the total is $(s+t+2+\Delta-s-1+\Delta-t-1)(st+1) = 2\Delta(st+1)$. Old points and lines have degree Δ, clones $t+1$, shadow lines $s+1$. The added pairs are anti-flags (with $\theta = \mathrm{id}$ no clone is on a shadow line); the blocks $\{ik, \dots, ik+k-1\}$, $i = 0, 1, \dots,$ run through $0, 1, \dots, k\sum_p \omega(p) - 1 = b(\Delta-s-1)-1$, so modulo b every residue is hit exactly $\Delta - s - 1$ times, and within a block the residues are distinct since $k \le b$. Adding flags does not increase distances (Lemma 2.2), and two non-collinear old points still share no line. □

Example 4.4 (Delorme's $D = 4$ construction [**?**, §8(a)]; the 2013 values)**.** In $W(3, q)$ take a spread Σ (a set of q^2+1 lines partitioning the points), a line $v \in \Sigma$, and $\mathcal{A}_1 = \{v\}$, $\mathcal{A}_0 = \Sigma \cup \{\text{points of } v\}$. (L1): the elements on v are its points; a point of v meets only v in $\mathcal{A}_1$, and a line of $\Sigma \setminus \{v\}$ meets nothing in $\mathcal{A}_1$. (L0): a point off v is on exactly one line of Σ and on no other line of $\mathcal{A}_0$; a line not in Σ has at most one point on v. The components of the substructure on $\mathcal{A}_0$ are the star formed by v and its points, and single lines, so (C_4) holds (Lemma 2.7). Theorem 2.6 gives maximum degree $q+m$, diameter 4 and $2(q+1)(q^2+1) + (m-1)\big((q^2+1) + (q+1) + m\big)$ elements. For $m = 2$ this is $2q^3 + 3q^2 + 3q + 6$: 346, 1734, 3064, 4946 at $q = 5, 9, 11, 13$; for $m = 3$, $q = 13$ it is $4760 + 2 \cdot 187 = 5134$. These are the 2013 values at $\Delta = 7, 11, 13, 15, 16$.

Remark 4.5 (a limit for shadow lines on $Q(4, q)$). Let q be odd, $\mathcal{O}$ an ovoid of $Q(4, q)$ (a set of q^2+1 points meeting every line once), $\omega = \mathbf{1}_{\mathcal{O}}$, and suppose the maximum degree is $q+2$. Then at most $q^2 - q + 2$ shadow lines are possible. Indeed every old point and every clone serves at most one shadow line, so a point off $\mathcal{O}$ is on at most one entry of $\mathcal{M}$ and a point of $\mathcal{O}$ on at most two. Suppose some $o \in \mathcal{O}$ is on two entries. They are distinct lines: if both were copies of one line N, each of the q points of N other than o is off $\mathcal{O}$, has no clone, and would be the stand-in of both entries, giving it degree $q+3$. Each other entry L misses o and so has a unique point y collinear with o; $y \notin \mathcal{O}$ (points of an ovoid are pairwise non-collinear); the line oy is not an entry of $\mathcal{M}$ (otherwise $y \notin \mathcal{O}$ would be on two entries), so it is one of the $q-1$ lines on o other than the two entries through o, and y is one of its q points other than o; distinct L give distinct y. So $|\mathcal{M}| \le 2 + q(q-1)$. Otherwise every point is on at most one entry, $\mathcal{M}$ is a partial spread, and $|\mathcal{M}| \le q^2 - q + 1$ (Tallini [**?**]; see also [**?**, §1]).

Example 4.6 ($(7, 4) \ge 359$)**.** In $Q(4, 5)$ take one clone of each point of an ovoid $\mathcal{O}$ that is an elliptic quadric section ($\omega = \mathbf{1}_{\mathcal{O}}$), and 21 shadow lines with stand-ins. Theorem 4.2 gives diameter 4, maximum degree 7, and $2 \cdot 6 \cdot 26 + 26 + 21 = 359$ elements. Appendix **??** lists the 21 lines and the stand-ins in coordinates, so that the conditions of Theorem 4.2(b) can be checked by hand. Every ovoid of $Q(4, p)$, p prime, is an elliptic quadric [**?**], so the choice of ovoid does not matter. By Remark 4.5 at most $q^2 - q + 2 = 22$ shadow lines are possible in this setting; 21 is the maximum for this family (an integer program, solved to optimality). This example is not a member of a proved infinite family. The best regular graph we know at $(7, 4)$ has 350 vertices (Corollary 4.8 at $q = 5$).

Example 4.6: the optimality of 21 comes from an integer-programming solver (HiGHS). Keep the sentence, or drop it as computer-dependent?

4.2. **The Ahrens–Szekeres quadrangle and two elliptic quadrics.**

Coordinates. Let q be odd. Let $W(q) = W(3, q)$ be the symplectic quadrangle of $\mathrm{PG}(3, q)$ for the alternating form

$$\beta(u, v) = u_0v_1 - u_1v_0 + u_2v_3 - u_3v_2,$$

and let $x = (1 : 0 : 0 : 0)$; then $x^\perp$ is the plane $X_1 = 0$. Payne's derived quadrangle $\mathrm{AS}(q) = P(W(q), x)$ [**?**], [**?**, 3.1.4] has

- *points*: the q^3 points $(a : 1 : b : c)$, $a, b, c \in \mathbb{F}$ (the points off $x^\perp$); we write (a, w) with $w = (b, c)$;
- *lines of type (i) (fibres)*, q^2 of them: $\{(a_0 + \mu : 1 : w_0) : \mu \in \mathbb{F}\}$, the line $\langle x, (a_0 : 1 : w_0)\rangle$ minus x;
- *lines of type (ii)*, $q^2(q+1)$ of them: for a point $(a_0 : 1 : w_0)$ and a direction $d = (d_1 : d_2) \in \mathrm{PG}(1, q)$,

$$\ell(a_0, w_0; d) = \{(a_0 + \lambda\rho : 1 : w_0 + \lambda d) : \lambda \in \mathbb{F}\}, \qquad \rho = \det(w_0, d) = b_0 d_2 - c_0 d_1,$$

the totally isotropic line through $(a_0 : 1 : w_0)$ and $(\rho : 0 : d_1 : d_2)$, minus the latter point.

(Check: $\beta((a_0, 1, b_0, c_0), (\rho, 0, d_1, d_2)) = -\rho + b_0 d_2 - c_0 d_1 = 0$. A point off $x^\perp$ is on $q+1$ totally isotropic lines, none through x, each meeting $x^\perp$ in one point $(\rho : 0 : d)$, $d \neq 0$; so each point of $\mathrm{AS}(q)$ is on one fibre and on $q+1$ lines of type (ii), and every line has q points.) $\mathrm{AS}(q)$ is a GQ of order $(q-1, q+1)$ [**?**], [**?**, 3.1.4, 3.2.6]; for q odd it is the Ahrens–Szekeres quadrangle [**?**]. The fibres form a spread.

Fix a non-square $\varepsilon \in \mathbb{F}$ and put $N(b, c) = b^2 - \varepsilon c^2$, an anisotropic form ($N(w) = 0$ only for $w = 0$). For $\kappa \in \mathbb{F}^*$ let $\mathcal{E}_\kappa$ be the quadric $X_0 X_1 = \kappa N(X_2, X_3)$. It is elliptic (a hyperbolic pair plus an anisotropic binary form), passes through x with tangent plane $x^\perp$, and $\mathcal{E}_\kappa \setminus \{x\} = \{(\kappa N(w) : 1 : w) : w \in \mathbb{F}^2\}$ is a set of q^2 points of $\mathrm{AS}(q)$, one on each fibre.

Theorem 4.7 (a weighted 2-ovoid of $\mathrm{AS}(q)$)**.** *Let q be odd, ε a non-square and $\kappa_1\kappa_2 = 1/(4\varepsilon)$. Every line of $\mathrm{AS}(q)$ meets $\mathcal{E}_{\kappa_1} \setminus \{x\}$ and $\mathcal{E}_{\kappa_2} \setminus \{x\}$ in exactly two points in total (a point in both counted twice). The two quadrics meet only in x and $e = (0 : 1 : 0 : 0)$. So $\varpi = \mathbf{1}_{\mathcal{E}_{\kappa_1} \setminus x} + \mathbf{1}_{\mathcal{E}_{\kappa_2} \setminus x}$ is a weighted 2-ovoid of $\mathrm{AS}(q)$, with $\varpi(e) = 2$ and $\varpi \in \{0, 1\}$ elsewhere.*

The choice $\kappa_1 = \frac{1}{2}$, $\kappa_2 = \frac{1}{2\varepsilon}$ gives the two quadrics

$$\mathcal{E}^{(1)} : \ 2X_0X_1 = X_2^2 - \varepsilon X_3^2, \qquad \mathcal{E}^{(2)} : \ 2\varepsilon X_0X_1 = X_2^2 - \varepsilon X_3^2.$$

Corollary 4.8. *For every odd prime power q, adjoining to $\mathrm{AS}(q)$ one clone of each point of $\mathcal{E}^{(1)} \setminus \{x\}$ and one of each point of $\mathcal{E}^{(2)} \setminus \{x\}$ (so e gets two clones) gives a $(q+2)$-regular incidence structure of diameter 4 with $2q^2(q+2)$ elements. Hence $n_b(q+2, 4) \geq 2q^2(q+2)$, attained by a regular graph.*

New vertices. For $P = (\kappa_i N(w) : 1 : w) \in \mathcal{E}_{\kappa_i} \setminus \{x\}$ ($i = 1, 2$, $w \in \mathbb{F}^2$), the clone $P^{(i)}$ is adjacent exactly to the $q+2$ lines of $\mathrm{AS}(q)$ through P: the fibre $\{(\kappa_i N(w) + \mu : 1 : w) : \mu \in \mathbb{F}\}$ and, for each $d \in \mathrm{PG}(1, q)$, the line $\ell(\kappa_i N(w), w; d)$. The two clones of e are adjacent to the same $q+2$ lines. Changed old neighbourhoods: the fibre over w_0 gains the clones of $(\kappa_1 N(w_0) : 1 : w_0)$ and $(\kappa_2 N(w_0) : 1 : w_0)$; the line $\ell(a_0, w_0; d)$ gains the clones of its points on $\mathcal{E}_{\kappa_1}$ and on $\mathcal{E}_{\kappa_2}$, which correspond to the roots λ of the quadratic (4.1) below with $\kappa = \kappa_1, \kappa_2$ (two clones in total). Old points are unchanged.

Proof of Theorem 4.7. Fibres. The fibre over w_0 meets $\mathcal{E}_\kappa$ only at $\mu = \kappa N(w_0) - a_0$: one point of each quadric.

Lines of type (ii). Let $B = N(w_0 + d) - N(w_0) - N(d) = 2(b_0 d_1 - \varepsilon c_0 d_2)$. The point with parameter λ of $\ell(a_0, w_0; d)$ lies on $\mathcal{E}_\kappa$ if and only if $a_0 + \lambda\rho = \kappa N(w_0 + \lambda d)$, that is,

$$\kappa N(d)\lambda^2 + (\kappa B - \rho)\lambda + \kappa N(w_0) - a_0 = 0. \tag{4.1}$$

As $N(d) \neq 0$ this is a quadratic, with $1 + \chi(\mathrm{disc}_\kappa)$ roots. A direct expansion gives $B^2 - 4N(w_0)N(d) = 4\varepsilon(b_0 d_2 - c_0 d_1)^2 = 4\varepsilon\rho^2$, hence

$$\mathrm{disc}_\kappa = (\kappa B - \rho)^2 - 4\kappa N(d)(\kappa N(w_0) - a_0) = 4\kappa N(d)\,(a_0 - a_\kappa^*), \qquad a_\kappa^* = -\frac{4\varepsilon\kappa\rho^2 - 2B\rho + \rho^2/\kappa}{4N(d)}.$$

Then $a^*_{\kappa_1} - a^*_{\kappa_2} = -\frac{\rho^2}{4N(d)}(\kappa_1-\kappa_2)\big(4\varepsilon - \frac{1}{\kappa_1\kappa_2}\big) = 0$; call the common value a^*. The line meets the two quadrics in

$$\big(1+\chi(\kappa_1)\chi(N(d)(a_0-a^*))\big) + \big(1+\chi(\kappa_2)\chi(N(d)(a_0-a^*))\big) = 2$$

points, because $\chi(\kappa_1)\chi(\kappa_2) = \chi(1/(4\varepsilon)) = \chi(\varepsilon) = -1$, so $\chi(\kappa_1)+\chi(\kappa_2) = 0$. (If $a_0 = a^*$, the line is tangent to both quadrics and meets each in one point.)

Intersection. $\kappa_1 \neq \kappa_2$, since $\kappa_1^2 = 1/(4\varepsilon)$ is impossible for a non-square ε. So $\kappa_1 N(w) = \kappa_2 N(w)$ forces $N(w) = 0$, $w = 0$, and the common points are x and e. □

Proof of Corollary 4.8. Theorem 4.2 and Corollary 4.3 with $(s,t) = (q-1, q+1)$, $\omega = \varpi$ of Theorem 4.7 and $\mathcal{M} = \emptyset$: every line has q points and gains exactly two clones, every old point is on $q+2$ lines, and every clone is on the $q+2$ lines of its point. The order is $q^3+q^2(q+2)+2q^2 = 2q^2(q+2)$. □

Remark 4.9. Weights are necessary: AS(3) has no 1- or 2-ovoid and AS(5) has no m-ovoid for $m = 1,\dots,4$ (computer-checked by integer programming). No choice of clones and shadow lines on AS(q) does better at $\Delta = q+2$: every point of AS(q) is already on $t+1 = q+2$ lines, so no shadow line can be added, and each line has q points and takes at most 2 clones, so $\sum_p \omega(p) \le 2q^2$. Corollary 4.8 beats Delorme's $2q^3+3q^2+3q+6$ by $q^2-3q-6 > 0$ for $q \ge 5$.

Cells. $(11,4)$: 1782 (2013: 1734); $(13,4)$: 3146 (3064); $(15,4)$: 5070 (4946). All regular.

4.3. **Singer 2-ovoids of $W(3,q)$.**

The cyclic model. Let q be odd, $\mathbb{F} = \mathrm{GF}(q) \subset K = \mathrm{GF}(q^2) \subset E = \mathrm{GF}(q^4)$, and g a primitive element of E. As E is a 4-dimensional $\mathbb{F}$-space, $\mathrm{PG}(3,q) = E^*/\mathbb{F}^*$ (any $\mathbb{F}$-basis of E gives homogeneous coordinates); the point $\langle g^i\rangle$ has index i modulo $(q+1)(q^2+1)$. Write $\bar{y} = y^{q^2}$ (the involution of E/K), $k' = k^q$ for $k \in K$, $\mathrm{N}(y) = y\bar{y} \in K$ and $\mathrm{Tr} = \mathrm{Tr}_{K/\mathbb{F}}$. Let $d = g^{(q^2+1)/2}$; then $\bar{d} = -d$, and $h = d^2 = g^{q^2+1}$ generates K^*. The form

$$b(y,z) = \mathrm{Tr}_{E/\mathbb{F}}(d\,y\bar{z}) = \mathrm{Tr}\big(d(y\bar{z} - \bar{y}z)\big)$$

is alternating and non-degenerate; $W(3,q)$ is its symplectic quadrangle. Multiplication by g^{q+1} multiplies b by $g^{(q+1)(q^2+1)} \in \mathbb{F}^*$, so it is a collineation of $W(3,q)$. Its point orbits are the classes

$$C_a = \{\langle g^i\rangle :\ i \equiv a \pmod{q+1}\}, \qquad |C_a| = q^2+1,$$

and $\langle y\rangle \in C_a$ if and only if $\mathrm{N}(y) \in h^a\mathbb{F}^*$ (as $\mathrm{N}(g^i) = h^i$ and $K^*/\mathbb{F}^*$ is cyclic of order $q+1$, generated by h). These are Ebert's elliptic quadrics partitioning $\mathrm{PG}(3,q)$ [**?**].

The K-subspaces yK ($y \in E^*$) are lines of $\mathrm{PG}(3,q)$ (the regular spread), with points $\langle g^{i+(q^2+1)j}\rangle$, $j = 0,\dots,q$. They are totally isotropic: for $k_1, k_2 \in K$ we have $\overline{yk_i} = \bar{y}k_i$ (as $\bar{k} = k^{q^2} = k$), so $b(yk_1, yk_2) = \mathrm{Tr}\big(d\,\mathrm{N}(y)(k_1k_2 - k_1k_2)\big) = 0$. So $\Sigma = \{yK\}$ is a spread of $W(3,q)$ with q^2+1 lines.

Proposition 4.10 (Singer 2-ovoids)**.** *Let q be odd.*

(a) Every line of $\mathrm{PG}(3,q)$ meets each class C_a in at most two points.

(b) Every totally isotropic line of $W(3,q)$ meets $C_a \cup C_{q-a}$ in exactly two points.

So the $(q+1)/2$ sets $C_a \cup C_{q-a}$, $0 \le a \le (q-1)/2$, are 2-ovoids of $W(3,q)$ partitioning its points. (The pairing $a \leftrightarrow q-a$ has no fixed point, as $2a \equiv -1 \pmod{q+1}$ is impossible for $q+1$ even.)

Proof. Let $L = \langle y,z\rangle_{\mathbb{F}}$ and $u = sy+tz$ ($s,t \in \mathbb{F}$). Then $\mathrm{N}(u) = A_1s^2 + \tau st + A_2t^2$ with $A_1 = \mathrm{N}(y)$, $A_2 = \mathrm{N}(z)$, $\tau = y\bar{z} + \bar{y}z$, all in K. Put $\theta = y\bar{z} - \bar{y}z$. As $\bar{\theta} = -\theta$, $\theta/d \in K$; write $\theta = dk_0$ with $k_0 \in K$. Then

$$\mathfrak{d} := \tau^2 - 4A_1A_2 = \theta^2 = hk_0^2, \qquad b(y,z) = \mathrm{Tr}(hk_0).$$

Fix $\varkappa \in K^*$ with $\varkappa' = -\varkappa$. For $\mu \in K^*$ put $\nu = \varkappa/\mu$; the kernel of $k \mapsto \mathrm{Tr}(\nu k)$ is the $\mathbb{F}$-subspace $\mu\mathbb{F}$. So $\langle u\rangle$ lies in $\widetilde{C}_\mu := \{\langle u\rangle : \mathrm{N}(u) \in \mu\mathbb{F}^*\}$ (thus $C_a = \widetilde{C}_{h^a}$) if and only if $(s:t)$ is a zero of the binary form

$$f_\nu(s,t) = \mathrm{Tr}(\nu A_1)s^2 + \mathrm{Tr}(\nu\tau)st + \mathrm{Tr}(\nu A_2)t^2.$$

(a) Suppose $f_\nu \equiv 0$. Then $A_1, \tau, A_2 \in \mu\mathbb{F}$. If $\mathfrak{d} \neq 0$, then $hk_0^2 = \mathfrak{d} \in \mu^2\mathbb{F}^*$, so $h \in (K^*)^2\mathbb{F}^* = (K^*)^2$ (every element of $\mathbb{F}^*$ is a square in K, since $q-1$ divides $(q^2-1)/2$); but h generates K^*, which has even order. If $\mathfrak{d} = 0$, then $\mathrm{N}(sy+tz) = \mu(\alpha_0 s^2 + \beta_0 st + \gamma_0 t^2)$ with $\alpha_0, \beta_0, \gamma_0 \in \mathbb{F}$, $\beta_0^2 = 4\alpha_0\gamma_0$ and $\alpha_0 \neq 0$ (as $A_1 \neq 0$), which vanishes at $(s:t) = (-\beta_0 : 2\alpha_0)$: impossible, since $\mathrm{N}(u) \neq 0$ for $u \neq 0$. So $f_\nu \not\equiv 0$, and it has $1 + \chi(\mathrm{disc}(\nu)) \leq 2$ projective zeros, where $\mathrm{disc}(\nu) = \mathrm{Tr}(\nu\tau)^2 - 4\mathrm{Tr}(\nu A_1)\mathrm{Tr}(\nu A_2)$.

(b) Expanding with $\mathrm{Tr}(\nu k) = \nu k + \nu' k'$,

$$\mathrm{disc}(\nu) = \mathrm{Tr}(\nu^2\mathfrak{d}) + \nu\nu'\Psi, \qquad \Psi = 2\tau\tau' - 4A_1A_2' - 4A_1'A_2 \in \mathbb{F}.$$

If L is totally isotropic, then $\mathrm{Tr}(hk_0) = 0$, so $hk_0 \in \varkappa\mathbb{F}$ and $\eta := h\mathfrak{d} = (hk_0)^2 \in \varkappa^2\mathbb{F} \subseteq \mathbb{F}$. Then $\mathrm{Tr}(h^2\nu'^2\mathfrak{d}) = \eta(h\nu'^2 + h'\nu^2) = hh'\,\mathrm{Tr}(\nu^2\mathfrak{d})$, and since $(h\nu')(h\nu')' = hh'\nu\nu'$,

$$\mathrm{disc}(h\nu') = \mathrm{N}_{K/\mathbb{F}}(h)\,\mathrm{disc}(\nu).$$

$\mathrm{N}_{K/\mathbb{F}}(h) = h^{q+1}$ generates $\mathbb{F}^*$, so it is a non-square and $\chi(\mathrm{disc}(h\nu')) = -\chi(\mathrm{disc}(\nu))$. The class belonging to $h\nu'$ is $\varkappa/(h\nu') = -\mu'/h \equiv (h\mu)^{-1}$ modulo $\mathbb{F}^*$ (using $\mu^{q+1} \in \mathbb{F}^*$); in indices, $a \mapsto -a-1 \equiv q-a \pmod{q+1}$. Scaling ν by $\mathbb{F}^*$ multiplies disc by a square. So L meets $C_a \cup C_{q-a}$ in $(1+\chi(\mathrm{disc}(\nu))) + (1-\chi(\mathrm{disc}(\nu))) = 2$ points. □

Remark 4.11 (literature). For q odd, $W(3,q)$ has m-ovoids for every even m and $(q+1)/2$-ovoids [?], and [?] states, citing [?], that $W(3,p^e)$ (p odd) has a partition into 2-ovoids. Cossidente, Culbert, Ebert and Marino [?] construct m-ovoids of $W(3,q)$ with Singer-type stabilisers. We have not checked whether the 2-ovoids $C_a \cup C_{q-a}$ coincide with those; Proposition 4.10 is very likely known, and we claim only the short proof. The proof uses only that d^2 is a non-square in K: for any δ_0 with $\bar{\delta}_0 = -\delta_0$, $C_a \cup C_b$ is a 2-ovoid of the form $\mathrm{Tr}_{E/\mathbb{F}}(\delta_0 y\bar{z})$ when $h^{a+b}\delta_0^2 \in \mathbb{F}^*$.

Proposition 4.10: do you know whether the 2-ovoids $C_a \cup C_{q-a}$ are those of [BLP09], [CCEM08] or [FWX19]? If so, cite and shorten.

Corollary 4.12. *Let q be odd, $0 \leq a \leq (q-1)/2$, $\mathcal{O} = C_a \cup C_{q-a}$ and Σ the regular spread $\{yK\}$. Adjoin to $W(3,q)$*

- *one clone P' of each point $P \in \mathcal{O}$, incident with the $q+1$ totally isotropic lines on P (the lines $\langle P, Z\rangle$ with $b(P,Z) = 0$);*
- *two shadow lines L', L'' for each $L \in \Sigma$, each incident with the $q+1$ points of L.*

The result is bipartite, has diameter 4, maximum degree $q+3$ (old points and lines have degree $q+3$, clones and shadow lines $q+1$) and $2(q^2+1)(q+3)$ elements. Joining clone c_i to shadow lines y_{2i}, y_{2i+1} (indices modulo $2(q^2+1)$, any numbering) makes it $(q+3)$-regular with diameter 4. Hence $n_b(q+3,4) \geq 2(q^2+1)(q+3)$, attained by a regular graph.

Proof. $W(3,q)$ is a GQ of order (q,q). By Proposition 4.10(b), $\omega = \mathbf{1}_{\mathcal{O}}$ is a 2-ovoid ($= \Delta - s - 1$ for $\Delta = q+3$), and $\mathcal{M} = 2\Sigma$ is a 2-cover (every point is on exactly one spread line). Apply Corollary 4.3 with $k = 2$. □

This is the point–line dual of clones of an ovoid of $Q(4,q)$ with a dual 2-cover; written in $W(3,q)$ it needs no Klein correspondence. More generally, j copies of the spread and $j/2$ copies of the 2-ovoid give $n_b(q+1+j,4) \geq 2(q^2+1)(q+1+j)$ for even j.

Corollary 4.13 (all degrees)**.** *For every prime power q and every $\Delta \geq q+1$ there is a Δ-regular incidence structure of diameter 4 with $2\Delta(q^2+1)$ elements if q is even or $\Delta-q-1$ is even, and with $2\Delta q^2$ elements otherwise. In particular $n_b(\Delta,4) \geq 2q^2\Delta$ for every prime power $q \leq \Delta-1$.*

Proof. Put $j = \Delta - q - 1$. In the first two cases we apply Corollary 4.3 in $W(3,q)$ ($s=t=q$, so $\Delta-s-1 = \Delta-t-1 = j$). If q is even, $W(3,q)$ has an ovoid $\mathcal{O}_0$ and a spread Σ [**?**, 3.4.1]; take $\omega = j\mathbf{1}_{\mathcal{O}_0}$ and $\mathcal{M} = j\Sigma$. If q is odd and j even: $\omega = (j/2)\mathbf{1}_{\mathcal{O}}$ with $\mathcal{O}$ a 2-ovoid of Proposition 4.10, and $\mathcal{M} = j\Sigma$. If q is odd and j odd, then $\Delta \geq q+2$, and we apply Corollary 4.3 in $\mathrm{AS}(q)$ ($s = q-1$, $t = q+1$, so $\Delta-s-1 = \Delta-q$ and $\Delta-t-1 = \Delta-q-2$), with $(\Delta-q)/2$ copies of the weighted 2-ovoid of Theorem 4.7 and $\Delta-q-2$ copies of the fibre spread. □

Cells. $(16,4)$: $q=13$, $5440 = 2\cdot 170\cdot 16$ (2013: 5134), regular. In the range $\Delta \leq 16$ this is the only open cell with $\Delta-3$ an odd prime power. Without stand-ins, 5440 is the most a structure of clones and shadow lines on $W(3,13)$ or $Q(4,13)$ of maximum degree 16 can have (each line takes at most 2 clones and each point lies on at most 2 shadow lines).

5. Diameter 6: whiskers in the split Cayley hexagon

Throughout Sections 5.1–5.4, q is odd; even q is treated in Section 5.5. We use the model of $H(q)$ in Section 2.

5.1. **Elliptic hyperplane sections and Hermitian spreads.** Let $v = \langle e_1, e_6\rangle$. It is a line of $Q(6,q)$ (e_1, e_6 are singular and $b(e_1,e_6)=0$) and a line of $H(q)$ (no equation of (2.1) involves p_{16}). Fix $c \in \mathbb{F}$ with $-c$ a non-square. For $z=(\alpha,\beta)\in\mathbb{F}^2$ let

$$\Pi_z:\ X_0 + cX_4 + \alpha X_2 + \beta X_5 = 0, \qquad \Pi = \Pi_{(0,0)},$$

and let S_z be the set of lines of $H(q)$ contained in Π_z, $S = S_{(0,0)}$. Every Π_z contains v.

Lemma 5.1. *For every $z \in \mathbb{F}^2$:*

(a) $\Pi_z \cap Q(6,q)$ is an elliptic quadric $Q^-(5,q)$.
*(b) Every point of $H(q)$ in Π_z is on exactly one line of S_z. So $|S_z| = q^3+1$, the lines of S_z are pairwise opposite, and $v \in S_z$ (S_z is a Hermitian spread [**?**]).*
(c) A line of $H(q)$ not in S_z meets Π_z in exactly one point.
(d) The points of $H(q)$ at distance at most 3 from v are exactly the points of $Q(6,q)$ with $X_2 = X_5 = 0$; among them only the points of v lie in Π_z.
(e) $S_z \cap S_{z'} = \{v\}$ for $z \neq z'$.

Proof. (a) On Π_z substitute $X_0 = -cX_4 - \alpha X_2 - \beta X_5$: $X_0X_4 + X_1X_5 + X_2X_6 - X_3^2 = (X_1 - \beta X_4)X_5 + X_2(X_6 - \alpha X_4) - (X_3^2 + cX_4^2)$, two hyperbolic planes plus the anisotropic form $X_3^2 + cX_4^2$. So the form on Π_z is non-degenerate of Witt index 2.

(b) Let $y \in \Pi_z \cap Q(6,q)$. By (a), $\Pi_z \cap Q(6,q)$ contains no totally singular plane, so $\pi_y \not\subseteq \Pi_z$ and $\pi_y \cap \Pi_z$ is one line through y: the only line of $H(q)$ on y inside Π_z. $Q^-(5,q)$ has $(q+1)(q^3+1)$ points, so $|S_z| = q^3+1$. Two lines of S_z are not concurrent (uniqueness). If two were at distance 4, a line N of $H(q)$ would meet both in distinct points, so $N \subseteq \Pi_z$, $N \in S_z$, concurrent with them: a contradiction. So they are opposite.

(c) A line not in the hyperplane Π_z meets it in one point.

(d) Let $y = ae_1 + be_6$ be on v and yu a line of $H(q)$. Then $p_{12} = p_{34}$ gives $au_2 = 0$, $p_{65} = p_{30}$ gives $bu_5 = 0$, and $b(y,u) = 0$ gives $au_5 + bu_2 = 0$; so $u_2 = u_5 = 0$. Conversely $Q(6,q) \cap \{X_2 = X_5 = 0\}$ is $X_0X_4 = X_3^2$, a cone with vertex v over a conic, with $(q+1)+(q+1)q^2$ points: exactly the number of points at distance at most 3 from v ($q+1$ on v, and q more on each of the $q(q+1)$ other lines meeting v). If such a point is in Π_z, then $X_0 = -cX_4$ and $X_3^2 + cX_4^2 = 0$, so $X_0 = X_3 = X_4 = 0$ and the point is on v.

(e) If $\ell \in S_z \cap S_{z'}$ with $\ell \neq v$, then $(\alpha-\alpha')X_2 + (\beta-\beta')X_5 = 0$ on ℓ, so some point of ℓ has $X_2 = X_5 = 0$ and is at distance at most 3 from v by (d); but ℓ and v are opposite by (b), so every point of ℓ is at distance 5 from v. □

5.2. **Delorme's levels and whiskers.** Let $\mathcal{P}(v)$ be the points of v, $\mathcal{L}_v$ the q^2+q lines other than v that meet v, and $\mathcal{P}_\Pi$ the $(q+1)(q^3+1)$ points of $H(q)$ in Π. A point is *free* if it is not in Π and is at distance 5 from v; by Lemma 5.1(d) there are $(q+1)(q^4+q^2+1)-(q+1)-q^2(q+1)-q^3(q+1) = q^3(q^2-1)$ of them.

Definition 5.2. A *whisker set* is a set W of lines of $H(q)$, each opposite v and not in S, such that no free point is on two lines of W. It is *exact* if every free point is on exactly one line of W.

A line opposite v and not in S meets Π in one point, not on v (Lemma 5.1(c),(d)), and its other q points are free (every point of a line opposite v is at distance 5 from v). So $|W| \le q^3(q^2-1)/q = q^2(q^2-1)$, with equality if and only if W is exact.

Theorem 5.3 (doubling with whiskers)**.** *Let W be a whisker set and $m \ge 2$. Put*

$$\mathcal{A}_2 = \{v\}, \qquad \mathcal{A}_1 = \{v\} \cup \mathcal{P}(v) \cup S, \qquad \mathcal{A}_0 = \mathcal{A}_1 \cup \mathcal{P}_\Pi \cup \mathcal{L}_v \cup W.$$

Then $\mathcal{A}$ satisfies (L0), (L1) and (C_6). Hence $H(q)^{(m)}_{\mathcal{A}}$ is bipartite with maximum degree $q+m$, diameter 6, and

$$2(q+1)(q^4+q^2+1) + (m-1)\big(q^4+2q^3+q^2+2q+2+|W|+m(q^3+q+2)+m^2\big)$$

elements.

Proof. (L1), $i=1$: the elements on v are its points (in $\mathcal{A}_1$); a point of v is on one element of $\mathcal{A}_2$, a line of $S \setminus \{v\}$ on none. (L1), $i=0$: the elements incident with $\mathcal{A}_1$ are the points of v, the lines of $\mathcal{L}_v$ and the points of Π, all in $\mathcal{A}_0$; a line of $\mathcal{L}_v$ meets v in one point and contains no other point of $\mathcal{A}_1$; a point of $\mathcal{P}_\Pi \setminus \mathcal{P}(v)$ is on exactly one line of S (Lemma 5.1(b)) and not on v; a whisker has no point on v. (L0): a point outside $\mathcal{A}_0$ is off Π. If it is at distance 3 from v, it is on exactly one line of $\mathcal{L}_v$ (two would close a circuit of length at most 6) and on no whisker (a whisker through it would be at distance at most 4 from v). If it is free, it is on no line of $\mathcal{L}_v$ and on at most one whisker. A line outside $\mathcal{A}_0$ is not in Π (the lines in Π are those of S), so it meets Π in one point, its only point in $\mathcal{A}_0$. (C_6): a line of $\mathcal{L}_v$ has no point of Π other than its point on v (Lemma 5.1(d)), and a whisker has exactly one point of Π, not on v. So the components of the substructure on $\mathcal{A}_0$ are

$$\{v\} \cup \mathcal{P}(v) \cup \mathcal{L}_v \qquad \text{and} \qquad \{n\} \cup \mathcal{P}(n) \cup \{h \in W : h \text{ meets } n\} \quad (n \in S \setminus \{v\}),$$

trees of diameter 4 (the ball of radius 2 about v; a line, its points and the whiskers hanging from them). Lemma 2.7 gives (C_6), and Theorem 2.6 gives degree at most $q+1+(m-1)$, diameter at most 6 and the count, with $|\mathcal{A}_2| = 1$, $|\mathcal{A}_1| = (q^3+1)+(q+1)$ and $|\mathcal{A}_0| = (q^3+1)+(q^2+q)+|W|+(q+1)(q^3+1)$. The diameter is exactly 6: a free point p (level -1) has q^5 opposite points, and only $(q+1)(q^3+1) < q^5$ points lie in $\mathcal{A}_0$, so some point p' opposite p has level -1, and $d(p,p') \ge 6$ by projection. □

The neighbourhood of every version. Write x^w for the versions ($i,j,k \in [m]$). Every element of $H(q)$ appears once in Table 4. An element of level k has m^{k+1} versions.

Table 4: The neighbourhood of every version in Theorem 5.3.

element (level)	versions	neighbours of the version	degree
v (2)	v^{ijk}	p^{ij} for the $q+1$ points p of v	$q+1$

element (level)	versions	neighbours of the version	degree
point p of v (1)	p^{ij}	$v^{ij1},\dots,v^{ijm}$; L^i for the q lines $L\in\mathcal{L}_v$ on p	$q+m$
line $n\in S\setminus\{v\}$ (1)	n^{ij}	p^i for the $q+1$ points p of n	$q+1$
point $p\in\mathcal{P}_\Pi\setminus\mathcal{P}(v)$ (0)	p^i	$n^{i1},\dots,n^{im}$ for the line $n\in S$ on p; h^i for each whisker h on p; every other line on p	$q+m$
line $L\in\mathcal{L}_v$ (0)	L^i	$p^{i1},\dots,p^{im}$ for $p=L\cap v$; the q other points of L	$q+m$
whisker $h\in W$ (0)	h^i	p^i for $p=h\cap\Pi$; the q free points of h	$q+1$
point $r\notin\Pi$ (-1)	r	all m versions of the line of $\mathcal{L}_v\cup W$ on r (if any); every other line on r	$q+m$, or $q+1$ if r is free and on no wh
line $N\notin\mathcal{A}_0$ (-1)	N	all m versions of its point in Π (not on v); its q other points	$q+m$

5.3. **An exact whisker set for every odd q.** For a point X of $H(q)$ in Π, not on v, put

$$\nu(X)=\frac{X_2X_3-X_4X_5}{cX_2^2+X_5^2}.$$

The denominator vanishes only if $X_2=X_5=0$, that is (Lemma 5.1(d)) only on v. Let

$$G(a,b)=(c_0a+c_1b)(a^2+cb^2)-b^3,\qquad F(z)=\frac{G(\alpha,\beta)}{\alpha^2+c\beta^2}\quad(z=(\alpha,\beta)\neq0),$$

with $c_0,c_1\in\mathbb{F}$ chosen so that G has no zero on $\mathbb{F}^2\setminus\{0\}$.

Lemma 5.4. *Such c_0,c_1 exist for every odd q.*

Proof. $G(a,0)=c_0a^3$, so $c_0\neq0$ is needed. For $b\neq0$ and $t=a/b$, $G(a,b)=b^3\big((c_0t+c_1)(t^2+c)-1\big)$, and $t^2+c\neq0$. So G has a zero with $b\neq0$ if and only if (c_0,c_1) lies on one of the q lines $c_0t+c_1=1/(t^2+c)$, $t\in\mathbb{F}$, of the (c_0,c_1)-plane; each has $q-1$ points with $c_0\neq0$. The lines for $t=0$ and $t=1$ meet at $c_0=1/(1+c)-1/c=-1/(c(1+c))\neq0$ ($c\neq-1$, as $-c$ is a non-square). So these lines cover at most $q(q-1)-1$ of the $q(q-1)$ points with $c_0\neq0$. □

One valid choice for the q of the table ($\mathbb{F}_9=\mathbb{F}_3[i]$, $i^2=-1$):

q	3	5	7	9	11	13
c	1	2	1	$1+i$	1	2
(c_0,c_1)	(1,2)	(1,0)	(1,0)	$(i,2i)$	(1,3)	(1,1)

Theorem 5.5 (exact whiskers)**.** *Let q be odd, $-c$ a non-square, and c_0,c_1 as in Lemma 5.4. Then*

$$W=\big\{\ell\in S_z\setminus\{v\}:\ z\in\mathbb{F}^2\setminus\{0\},\ \nu(\ell\cap\Pi)=F(z)\big\}$$

is an exact whisker set: every free point lies on exactly one line of W, and $|W|=q^2(q^2-1)$.

Here $\ell\cap\Pi$ is one point not on v: $\ell\subseteq\Pi_z$ and $\ell\neq v$, so $\ell\notin S$ by Lemma 5.1(e), and ℓ is opposite v by Lemma 5.1(b). The proof uses a complex structure on each Π_z. Write a point of Π_z as $x=(x_1,\dots,x_6)$, meaning $(X_0,\dots,X_6)=(-cx_4-\alpha x_2-\beta x_5,x_1,\dots,x_6)$; on Π_z the quadratic form is $Q_z(x)=x_1x_5+x_2x_6-x_3^2-cx_4^2-\alpha x_2x_4-\beta x_4x_5$, with polar form b_z. Put $\mathrm{num}(y)=y_2y_3-y_4y_5$ and $\mathrm{den}(y)=cy_2^2+y_5^2$, and define the linear map

$$J_zx=\big(-\beta x_3+\alpha x_4-x_6,\ x_5,\ -cx_4-\beta x_5,\ x_3+\beta x_2,\ -cx_2,\ cx_1+\alpha\beta x_2+\alpha x_3+c\beta x_4+\beta^2x_5\big);$$

for $z = 0$, $J_0 x = (-x_6, x_5, -cx_4, x_3, -cx_2, cx_1)$.

Lemma 5.6 (complex structure)**.** *For every z:*

(a) $J_z^2 = -c$, $Q_z(J_z x) = cQ_z(x)$ *and* $b_z(x, J_z x) = 0$.
(b) The six differences $p_{12} - p_{34}$, $p_{54} - p_{32}$, $p_{20} - p_{35}$, $p_{65} - p_{30}$, $p_{01} - p_{36}$, $p_{46} - p_{31}$ *of the Grassmann coordinates of* $x \wedge J_z x$ *equal* $Q_z(x)$ *times* $1, 0, 0, -c, \alpha, -\beta$.
(c) For every point x of $Q(6,q) \cap \Pi_z$, the line $\langle x, J_z x\rangle$ is the line of S_z through x.
(d) $\operatorname{num}(sx + tJ_0 x) = (s^2 + ct^2)\operatorname{num}(x)$ *and* $\operatorname{den}(sx + tJ_0 x) = (s^2 + ct^2)\operatorname{den}(x)$. *So ν is constant on every line of $S \setminus \{v\}$.*

Proof. (a), (b) and (d) are polynomial identities with integer coefficients, verified by direct expansion. For instance, for $z = 0$: $Q_0(J_0 x) = cx_2x_6 + cx_1x_5 - c^2x_4^2 - cx_3^2 = cQ_0(x)$; $p_{12} = x_1x_5 + x_2x_6$ and $p_{34} = x_3^2 + cx_4^2$, so $p_{12} - p_{34} = Q_0(x)$; in (d), $\operatorname{num}(J_0 x) = c\operatorname{num}(x)$ and $\operatorname{den}(J_0 x) = c\operatorname{den}(x)$, and the polar forms $x_2y_3 + x_3y_2 - x_4y_5 - x_5y_4$ and $2cx_2y_2 + 2x_5y_5$ vanish at $y = J_0 x$. (c) J_z has no eigenvector ($-c$ is a non-square), so x and $J_z x$ span a line; if $Q_z(x) = 0$, then by (a) the whole line is singular and by (b) its Grassmann coordinates satisfy (2.1), so it is a line of $H(q)$ in Π_z through x, the one of Lemma 5.1(b). (d) By (c), a line $n \in S \setminus \{v\}$ is $\{sx + tJ_0 x\}$ for any of its points x, and $s^2 + ct^2 \neq 0$. □

Let p be a free point. By Lemma 5.1(d), $(p_2, p_5) \neq (0,0)$; as $p \notin \Pi$ we normalise $p_0 + cp_4 = 1$. Then $p \in \Pi_z$ if and only if z lies on the affine line

$$A_p = \{(\alpha, \beta) :\ \alpha p_2 + \beta p_5 = -1\} \subseteq \mathbb{F}^2 \setminus \{0\},$$

which has q points and direction $(p_5, -p_2)$. For $z \in A_p$ let $\ell_p(z)$ be the line of S_z on p and $X(z) = \ell_p(z) \cap \Pi$.

Lemma 5.7 (key identity)**.** *For $z = (\alpha, \beta) \in A_p$,*

$$\Phi_p(z) := \nu(X(z)) + \frac{\beta^3}{\alpha^2 + c\beta^2} = \frac{p_2p_3 - p_4p_5 + \beta p_2^2}{cp_2^2 + p_5^2}.$$

So Φ_p is affine on A_p, with increment $-p_2^3/(cp_2^2 + p_5^2)$ along $(p_5, -p_2)$.

Proof. Put $\Theta = cp_2^2 + p_5^2 \neq 0$ and $\gamma = -\alpha p_5 + c\beta p_2$. On Π_z the linear form $\Lambda(y) = y_0 + cy_4$ equals $-\alpha y_2 - \beta y_5$, and $\Pi = \{\Lambda = 0\}$. We have $\Lambda(p) = 1$ and $\Lambda(J_z p) = -\alpha p_5 + c\beta p_2 = \gamma$. By Lemma 5.6(c), $\ell_p(z) = \langle p, J_z p\rangle$, so $X(z) = \gamma p - J_z p$, with

$$X_2 = \gamma p_2 - p_5, \quad X_3 = \gamma p_3 + cp_4 + \beta p_5, \quad X_4 = \gamma p_4 - p_3 - \beta p_2, \quad X_5 = \gamma p_5 + cp_2.$$

Expanding, $\operatorname{den}(X) = (\gamma^2 + c)\Theta$ and $\operatorname{num}(X) = (\gamma^2 + c)(p_2p_3 - p_4p_5) + \beta(2\gamma p_2p_5 + cp_2^2 - p_5^2)$. Solving $\alpha p_2 + \beta p_5 = -1$, $-\alpha p_5 + c\beta p_2 = \gamma$ (determinant Θ) gives $\alpha = -(cp_2 + \gamma p_5)/\Theta$ and $\beta = (\gamma p_2 - p_5)/\Theta$, hence $\alpha^2 + c\beta^2 = (\gamma^2 + c)/\Theta \neq 0$ and $\beta^3/(\alpha^2 + c\beta^2) = \beta(\gamma p_2 - p_5)^2/((\gamma^2 + c)\Theta)$. The two β-terms add up to $\beta\big(2\gamma p_2p_5 + cp_2^2 - p_5^2 + (\gamma p_2 - p_5)^2\big)/((\gamma^2 + c)\Theta) = \beta p_2^2/\Theta$. Along $(p_5, -p_2)$, β changes by $-p_2$. □

Proof of Theorem 5.5. Let p be free. The lines of $\bigcup_{z \neq 0} S_z \setminus \{v\}$ on p are exactly the q lines $\ell_p(z)$, $z \in A_p$, and they are distinct (a common line would lie in $S_z \cap S_{z'} = \{v\}$); for the same reason each line of W belongs to a single S_z. As $F(z) = c_0\alpha + c_1\beta - \beta^3/(\alpha^2 + c\beta^2)$, we have $\ell_p(z) \in W$ if and only if $\Phi_p(z) - (c_0\alpha + c_1\beta) = 0$. By Lemma 5.7 the left side is affine on A_p with increment

$$-\frac{p_2^3}{\Theta} - (c_0p_5 - c_1p_2) = -\frac{G(p_5, -p_2)}{\Theta} \neq 0$$

along $(p_5, -p_2)$, so it vanishes at exactly one of the q points of A_p: p is on exactly one line of W. The lines of W are opposite v and not in S. Counting incidences between free points and W gives $|W| = q^3(q^2 - 1)/q$. □

5.4. **The bound.**

Theorem 5.8. *For every odd prime power q and every $m \geq 2$, the structure of Theorem 5.3 with the exact whisker set of Theorem 5.5 is bipartite, has maximum degree $q+m$ and diameter 6, and*

$$n_b(q+m,6) \geq 2(q+1)(q^4+q^2+1) + (m-1)\big(2q^4+2q^3+2q+2+m(q^3+q+2)+m^2\big).$$

In particular $n_b(q+2,6) \geq 2(q+1)(q^4+q^2+1) + 2q^4+4q^3+4q+10$ and $n_b(q+3,6) \geq 2(q+1)(q^4+q^2+1) + 4q^4+10q^3+10q+34$.

Proof. Theorems 5.3 and 5.5, with $|W| = q^4-q^2$. □

With W exact, every element outside $\mathcal{A}_0$ is incident with exactly one element of $\mathcal{A}_0$. For a doubling of a generalized n-gon of order (q,q) with N points, counting incidences gives $q|\mathcal{A}_0|+2k \leq 2N$, k the number of components of $\mathcal{A}_0$; here $k = q^3+1$ and $q(2q^4+2q^3+2q+2)+2(q^3+1) = 2N$, so the single-level part is as large as this counting bound allows for q^3+1 components.

Remark 5.9 (Delorme's construction; the 2013 values). $W = \emptyset$ is Delorme's doubling of a generalized hexagon [**?**, §8(b)]: he takes q^3+1 pairwise opposite elements (citing [**?**]; here the Hermitian spread S), puts one of them in $\mathcal{A}_2$, its neighbours and the others in $\mathcal{A}_1$, and their neighbours in $\mathcal{A}_0$, so $|\mathcal{A}_1| = q^3+q+2$ and $|\mathcal{A}_0| = q^4+2q^3+q^2+2q+2$, exactly as in Theorem 5.3 with $W = \emptyset$; at $q = 5$ he states $7812+912+2\cdot 132+4 = 8992$. At $m = 2$ this has $2(q+1)(q^4+q^2+1)+q^4+4q^3+q^2+4q+10$ elements: 8992, 142464, 374452, 842048 at $q = 5, 9, 11, 13$; at $q = 13$, $m = 3$, it has 884062. These are the 2013 values at $\Delta = 7, 11, 13, 15, 16$. The exact whiskers add $(m-1)q^2(q^2-1)$.

Worked counts. $q = 5$, $m = 2$: $H(5)$ has $2\cdot 6\cdot 651 = 7812$ elements, $|\mathcal{A}_0| = 1512$ (of which $|W| = 600$), $|\mathcal{A}_1| = 132$, $|\mathcal{A}_2| = 1$, so the order is $7812+1512+2\cdot 132+4 = 9592$. $q = 13$, $m = 3$: $H(13)$ has 804468 elements, $|\mathcal{A}_0| = 61544$ ($|W| = 28392$), $|\mathcal{A}_1| = 2212$, so the order is $804468+2(61544+3\cdot 2212+9) = 940846$.
Cells. $m = 2$: $(7,6)$ 9592 (2013: 8992), $(11,6)$ 148944 (142464), $(13,6)$ 388972 (374452), $(15,6)$ 870440 (842048). $m = 3$, $q = 13$: $(16,6)$ 940846 (884062). These graphs are not regular; see Section 5.6.

5.5. **Even q.** Let $q = 2^e$, Tr the absolute trace of $\mathbb{F}$, and fix $\delta \in \mathbb{F}$ with $\mathrm{Tr}(\delta) = 1$, so that $t^2+t+\delta$ is irreducible and $a^2+ab+\delta b^2$ is anisotropic. Now $Q(6,q)$ is $X_0X_4+X_1X_5+X_2X_6+X_3^2 = 0$, with nucleus e_3, and $H(q)$ is given by the same equations (2.1). Put

$$\Pi_z:\ X_3 = X_0+\delta X_4+\alpha X_2+\beta X_5 \qquad (z=(\alpha,\beta)\in\mathbb{F}^2), \qquad \Pi = \Pi_{(0,0)},$$

S_z the lines of $H(q)$ in Π_z, and $v = \langle e_1, e_6\rangle \subseteq \Pi_z$ as before.

Lemma 5.10. *Lemma 5.1 holds verbatim for even q with these Π_z.*

Proof. (a) On Π_z, $X_0X_4+X_1X_5+X_2X_6+X_3^2 = (X_0^2+X_0X_4+\delta^2X_4^2)+(X_1+\beta^2X_5)X_5+X_2(X_6+\alpha^2X_2)$. The first term is anisotropic ($\mathrm{Tr}(\delta^2) = \mathrm{Tr}(\delta) = 1$) and the polar form is non-degenerate on Π_z (which misses e_3), so $\Pi_z \cap Q(6,q)$ is an elliptic quadric $Q^-(5,q)$. (b), (c), (e) are proved as in Lemma 5.1. (d) The same three equations give $u_2 = u_5 = 0$; on $X_2 = X_5 = 0$ the quadric is the cone $X_0X_4 = X_3^2$, and in Π_z this becomes $X_0^2+X_0X_4+\delta^2X_4^2 = 0$, so $X_0 = X_4 = X_3 = 0$, a point of v. □

So Theorem 5.3 holds for even q without change. For $X \in \Pi$ not on v put

$$\nu(X) = \frac{X_2X_3+X_2X_4+X_4X_5}{\delta X_2^2+X_2X_5+X_5^2}, \quad G(a,b) = (c_0a+c_1b)(a^2+ab+\delta b^2)+b^3, \quad F(z) = \frac{G(\alpha,\beta)}{\alpha^2+\alpha\beta+\delta\beta^2}.$$

The denominator of ν is anisotropic and vanishes only on v.

Lemma 5.11. *There are $c_0, c_1 \in \mathbb{F}$ with G zero-free on $\mathbb{F}^2 \setminus \{0\}$ if and only if $q \geq 4$.*

Proof. $G(a,0) = c_0 a^3$ needs $c_0 \neq 0$. For $b \neq 0$ and $t = a/b$: $G = 0$ if and only if $c_0 t + c_1 = r(t) := 1/(t^2+t+\delta)$, a line in the (c_0, c_1)-plane with $q-1$ points having $c_0 \neq 0$. If $q \geq 4$ take $\varpi \notin \{0,1\}$; then $r(\varpi) \neq r(0)$ (as $\varpi^2 + \varpi \neq 0$), so the lines for $t = 0, \varpi$ meet at $c_0 = (r(0) + r(\varpi))/\varpi \neq 0$, and the q lines cover fewer than $q(q-1)$ points with $c_0 \neq 0$. If $q = 2$ then $\delta = 1$, $r(0) = r(1) = 1$, and the lines $c_1 = 1$ and $c_0 + c_1 = 1$ contain both points with $c_0 = 1$. □

On Π_z use coordinates $x = (x_0, x_1, x_2, x_4, x_5, x_6)$, with $X_3 = x_0 + \delta x_4 + \alpha x_2 + \beta x_5$, and

$$J_z x = \big((\delta+1)x_0 + \delta\alpha x_2 + \delta^2 x_4 + (\alpha+\delta\beta)x_5,\ \beta x_0 + x_1 + \alpha\beta x_2 + (\alpha+\delta\beta)x_4 + \beta^2 x_5 + x_6,\ x_2 + x_5,$$
$$x_0 + (\alpha+\beta)x_2 + \delta x_4 + \beta x_5,\ \delta x_2,\ (\alpha+\beta)x_0 + \delta x_1 + \alpha^2 x_2 + \delta\alpha x_4 + \alpha\beta x_5\big).$$

Lemma 5.12. *With Q the quadratic form and b its polar form: (a) $J_z^2 + J_z + \delta = 0$, $Q(J_z x) = \delta Q(x)$ and $b(x, J_z x) = Q(x)$; (b) the six differences of (2.1) for $x \wedge J_z x$ are $Q(x)$ times $1, 0, 0, \delta, \alpha, \beta$; (c) for every point x of $Q(6,q) \cap \Pi_z$, $\langle x, J_z x\rangle$ is the line of S_z on x; (d) with* num $= X_2X_3 + X_2X_4 + X_4X_5$ *and* den $= \delta X_2^2 + X_2X_5 + X_5^2$ *on Π,* num$(sx + tJ_0x) = (s^2 + st + \delta t^2)num(x)$, *and likewise for* den*; so ν is constant on every line of $S \setminus \{v\}$.*

Proof. (a), (b), (d) are polynomial identities over $\mathbb{F}_2[\delta, \alpha, \beta, x]$, verified by expansion. (c) J_z has no eigenvector since $t^2+t+\delta$ is irreducible; if $Q(x) = 0$ then $Q(sx+tJ_zx) = (s^2+st+\delta t^2)Q(x) = 0$ by (a), and by (b) the line satisfies (2.1), so it is the line of Lemma 5.10(b). In (d), $s^2+st+\delta t^2 \neq 0$ for $(s,t) \neq 0$. □

Let p be free; then $(p_2, p_5) \neq (0,0)$ by Lemma 5.10(d), and we normalise $p_0 + p_3 + \delta p_4 = 1$. With $\Lambda(y) = y_0 + y_3 + \delta y_4$ we have $\Pi = \{\Lambda = 0\}$ and $\Lambda = \alpha y_2 + \beta y_5$ on Π_z, so $p \in \Pi_z$ if and only if z lies on the affine line $A_p:\ \alpha p_2 + \beta p_5 = 1$ (direction (p_5, p_2)). For $z \in A_p$ let $\ell_p(z)$ be the line of S_z on p and $X(z) = \ell_p(z) \cap \Pi$.

Lemma 5.13 (key identity, characteristic 2)**.** *For $z = (\alpha, \beta) \in A_p$,*

$$\Phi_p(z) := \nu(X(z)) + \frac{\beta^3}{\alpha^2 + \alpha\beta + \delta\beta^2} = \frac{p_2p_3 + p_2p_4 + p_4p_5 + \beta p_2^2}{\delta p_2^2 + p_2p_5 + p_5^2},$$

which is affine on A_p with increment p_2^3/Θ along (p_5, p_2), $\Theta = \delta p_2^2 + p_2p_5 + p_5^2 \neq 0$.

Proof. $\Lambda(p) = 1$ and $\gamma := \Lambda(J_z p) = \alpha(p_2+p_5) + \delta\beta p_2$. By Lemma 5.12(c), $X(z) = \gamma p + J_z p$ (then $\Lambda(X) = 2\gamma = 0$); $X_2 = (\gamma+1)p_2 + p_5$ and $X_5 = \delta p_2 + \gamma p_5$. Expanding, den$(X) = (\gamma^2+\gamma+\delta)\Theta$ and num$(X) = (\gamma^2+\gamma+\delta)(p_2p_3+p_2p_4+p_4p_5) + \beta\big((\gamma+\delta+1)p_2^2 + p_5^2\big)$, with $\gamma^2+\gamma+\delta \neq 0$. Solving $\alpha p_2 + \beta p_5 = 1$, $\alpha(p_2+p_5) + \delta\beta p_2 = \gamma$ (determinant Θ): $\alpha = (\delta p_2 + \gamma p_5)/\Theta$ and $\beta = ((\gamma+1)p_2 + p_5)/\Theta$, so $\alpha^2 + \alpha\beta + \delta\beta^2 = (\gamma^2+\gamma+\delta)/\Theta$ and $\beta^3/(\alpha^2+\alpha\beta+\delta\beta^2) = \beta((\gamma+1)p_2+p_5)^2/((\gamma^2+\gamma+\delta)\Theta)$. The β-terms add to $\beta\big((\gamma+\delta+1) + (\gamma+1)^2\big)p_2^2/((\gamma^2+\gamma+\delta)\Theta) = \beta p_2^2/\Theta$. Along (p_5, p_2), β changes by p_2. □

Theorem 5.14 (exact whiskers, even $q \geq 4$)**.** *With c_0, c_1 as in Lemma 5.11, $W = \{\ell \in S_z \setminus \{v\} : z \neq 0,\ \nu(\ell \cap \Pi) = F(z)\}$ is an exact whisker set.*

Proof. As for Theorem 5.5: the lines of $\bigcup_{z \neq 0} S_z \setminus \{v\}$ on a free point p are the q distinct lines $\ell_p(z)$, $z \in A_p$, and $\ell_p(z) \in W$ if and only if $\Phi_p(z) + c_0\alpha + c_1\beta = 0$, since $F(z) = c_0\alpha + c_1\beta + \beta^3/(\alpha^2 + \alpha\beta + \delta\beta^2)$. The left side is affine on A_p with increment $p_2^3/\Theta + c_0p_5 + c_1p_2 = G(p_5, p_2)/\Theta \neq 0$, so it vanishes at exactly one point. □

Corollary 5.15. *Theorem 5.8 and Theorem 5.18 hold for every even $q \geq 4$. For $q = 2$ there is also an exact whisker set (of $12 = q^2(q^2-1)$ lines, given explicitly and computer-checked), so they hold for $q = 2$ as well.*

None of this changes a cell with $\Delta \le 16$, for any m: for $q \in \{2,4,8\}$ and $\Delta = q+m \le 16$, either $\Delta - 1$ is a prime power (a cell where the Moore bound is attained by $H(\Delta-1)$), or $\Delta \in \{7,11,13,15,16\}$, where Theorem 5.8 with the odd $q' = \Delta - 2$ (or $q' = 13$, $m = 3$ at $\Delta = 16$) gives much more (for instance 9592 against 4468 from $q = 4$, $m = 3$ at $\Delta = 7$).

Corollary 5.15, $q = 2$: the 12 whiskers are a computer-found set (its exactness is also proved in Lean). List them in an appendix, or drop the $q = 2$ case?

5.6. **Regular graphs of diameter 6.** The structure of Theorem 5.8 has equally many points and lines in $\mathcal{A}_0$ (exact whiskers force this), but $\mathcal{A}_1$ has q^3+1 lines and only $q+1$ points, and $\mathcal{A}_2$ is a line. So it has $(m-1)m(q^3-q+m)$ more lines than points. We delete repeated lines and then add anti-flags.

Lemma 5.16 (repeated lines)**.** *In $H(q)^{(m)}_{\mathcal{A}}$, for $n \in S \setminus \{v\}$ and $i \in [m]$, the m lines $n^{i1}, \dots, n^{im}$ all have the point set $\{p^i : p \in n\}$; for $i, j \in [m]$ the lines $v^{ij1}, \dots, v^{ijm}$ all have the point set $\{p^{ij} : p \in v\}$ (Table 4). So there are $mq^3 + m^2$ classes of m repeated lines.*

Lemma 5.17 (completion)**.** *Let a finite incidence structure have maximum degree at most Δ, every element of degree at least $\Delta - r$, and $\sum_{points}(\Delta - \deg) = \sum_{lines}(\Delta - \deg) =: T$. If $T > 2r\Delta$, there is a set J of anti-flags such that adding J makes it Δ-regular.*

Proof. Choose a multiset J of point–line pairs in which each element x occurs $\Delta - \deg x$ times (possible as the sums agree). Call an entry bad if it is a flag or a repeat of another entry, and take J with the fewest bad entries. If (p, L) is bad: L is joined (in $\mathrm{I} \cup J$, with multiplicity) to Δ points, each in at most r entries of J; likewise p. So at most $2r\Delta < T$ entries (p', L') have p' joined to L or L' joined to p, and some entry (p', L') has neither (it is not (p, L) itself). Replacing $(p, L), (p', L')$ by $(p, L'), (p', L)$ removes a bad entry and creates none. So the minimum is 0. □

Theorem 5.18. *For every odd prime power q and $m \ge 2$ there is a $(q+m)$-regular incidence structure of diameter at most 6 with*

$$2(q+1)\big(q^4+q^2+1+(m-1)(q^3+m+1)\big)$$

elements: $2(q+1)(q^4+q^3+q^2+4)$ for $m = 2$ and $2(q+1)(q^4+2q^3+q^2+9)$ for $m = 3$. For the five table instances below the order exceeds $M_b(q+m, 5)$, so the diameter is exactly 6.

Proof. Start from Theorem 5.8. Fix a set R of q lines of $S \setminus \{v\}$, and delete the lines v^{ijk} with $k \ge 2$ ($(m-1)m^2$ lines) and n^{ij} with $j \ge 2$ for $n \notin R$ ($(m-1)m(q^3-q)$ lines): in total $(m-1)m(q^3-q+m)$, the excess, so points and lines are now equally many. Each class of Lemma 5.16 keeps at least one line, so by Lemma 2.2(b) no distance between remaining elements increases. After the deletion the degrees are: p^{ij} ($p \in v$): $q+1$; p^i on a line $n \notin R$ of S: $q+1$; p^i on $n \in R$: $q+m$; v^{ij1}, n^{i1} ($n \notin R$), n^{ij} ($n \in R$) and h^i ($h \in W$): $q+1$; all others as in Table 4 ($q+m$). So every deficit is at most $r = m-1$. Points and lines are equally many, so both deficit sums equal $\Delta|\mathcal{P}| - |E| = \Delta|\mathcal{L}| - |E|$ ($\Delta = q+m$, E the set of flags), as Lemma 5.17 requires; and their common value T is at least the deficit of the $m|W|$ whisker versions: $T \ge (m-1)mq^2(q^2-1) > 2(m-1)(q+m)$. Lemma 5.17 completes the structure; the added flags join a point to a line, so the graph stays bipartite, and they do not increase distances. The number of points is $(q+1)(q^4+q^2+1)+(m-1)\big((q+1)(q^3+1)+m(q+1)\big)$. □

Neighbourhoods. Every neighbourhood is that of Table 4, minus the deleted lines listed in the proof, plus the completing anti-flags: exactly $m-1$ of them at each element of degree $q+1$, none elsewhere. The completing anti-flags are given by Lemma 5.17, not by a formula; neither the maximum degree nor the diameter depends on the choice.

Theorem 5.18: the completing anti-flags are only shown to exist (as in the Lean proof). If you want every neighbourhood written out, we need an explicit completion (for instance a formula on the deficient elements); this is open work.

Cells (regular graphs). $(7,6)$ 9348, $(11,6)$ 147500, $(13,6)$ 386328, $(15,6)$ 866068, $(16,6)$ 927724 ($q = 13$, $m = 3$). Each beats the 2013 value of its cell (8992, 142464, 374452, 842048, 884062), and Delorme's graphs behind those values are not regular (their point side is larger).

References